

Optimized Virtual Orbital Space (OVOS) for Quantum Computing

Master's Thesis Defense

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Motivation: The Quantum Bottleneck

- **The Challenge:** Post-HF methods (MP2, CCSD) scale steeply $\mathcal{O}(O^2V^2)$ due to the virtual orbital space V .
- **NISQ Constraint:** Near-term quantum devices (VQE) are limited by qubit count and circuit depth.
- **The Insight:** Most virtual orbitals contribute negligibly to electron correlation.
- **Thesis Goal:** Implement the *Optimized Virtual Orbital Space (OVOS)* method (Adamowicz & Bartlett, 1987) to reduce virtual orbitals while preserving correlation energy. Benchmark these orbitals for VQE performance.

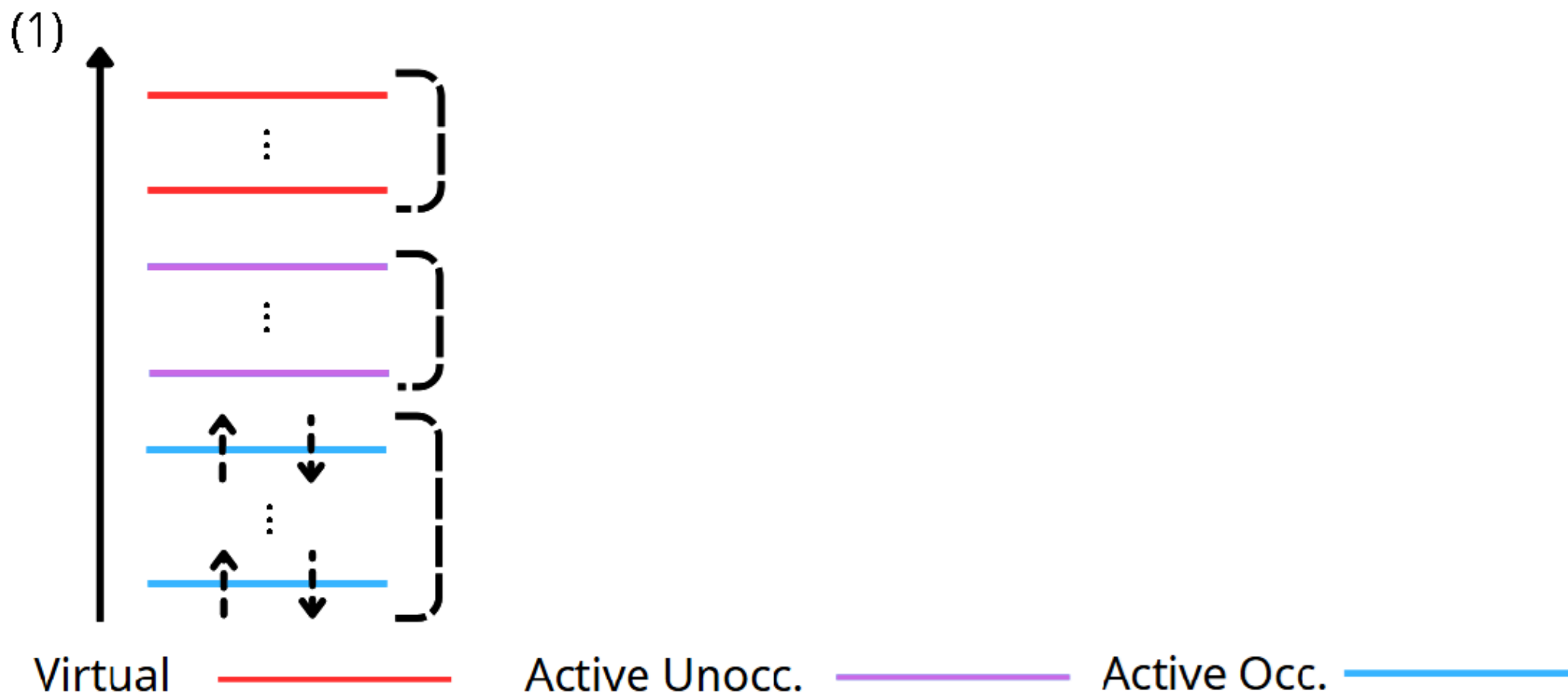
Theory

Partitioning - Optimisation



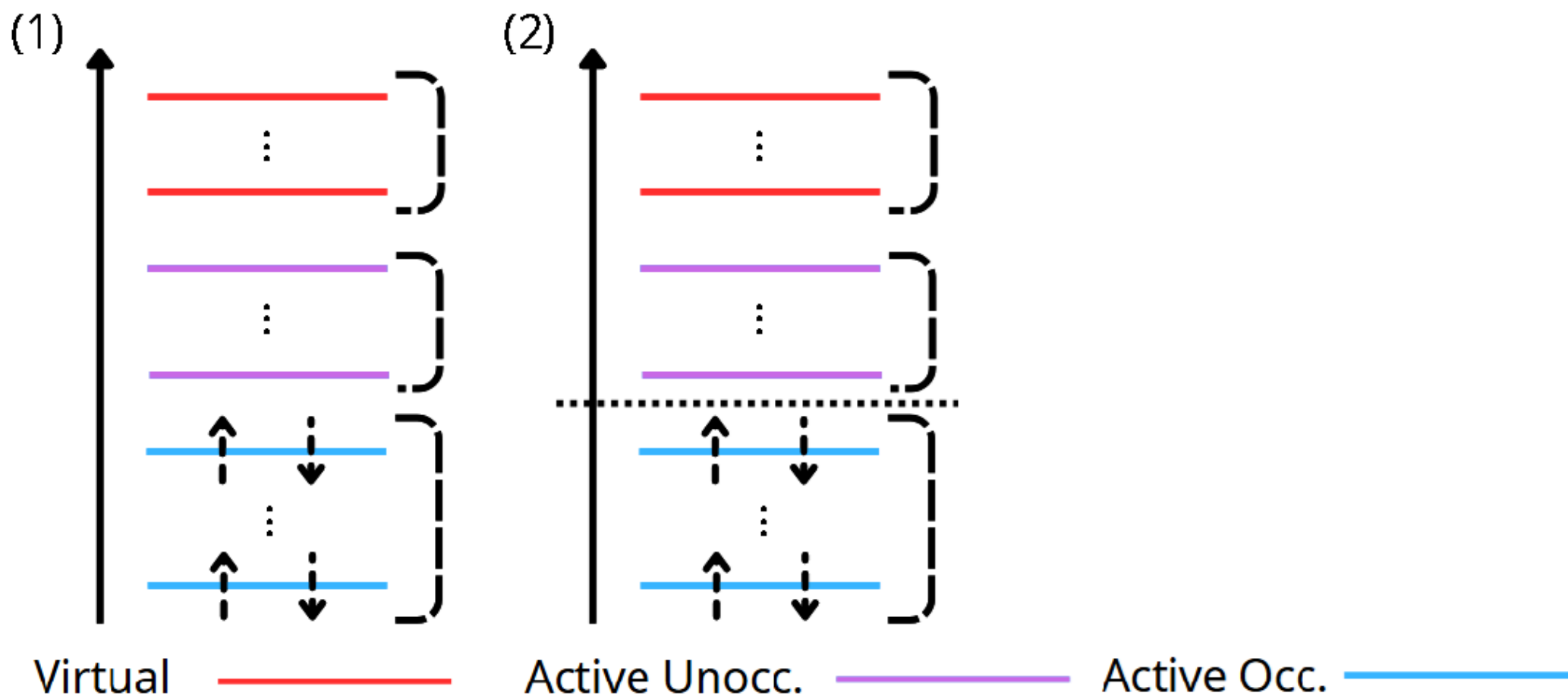
Theory: Partitioning

- **Index:** The virtual orbital space is partitioned into three subspaces:



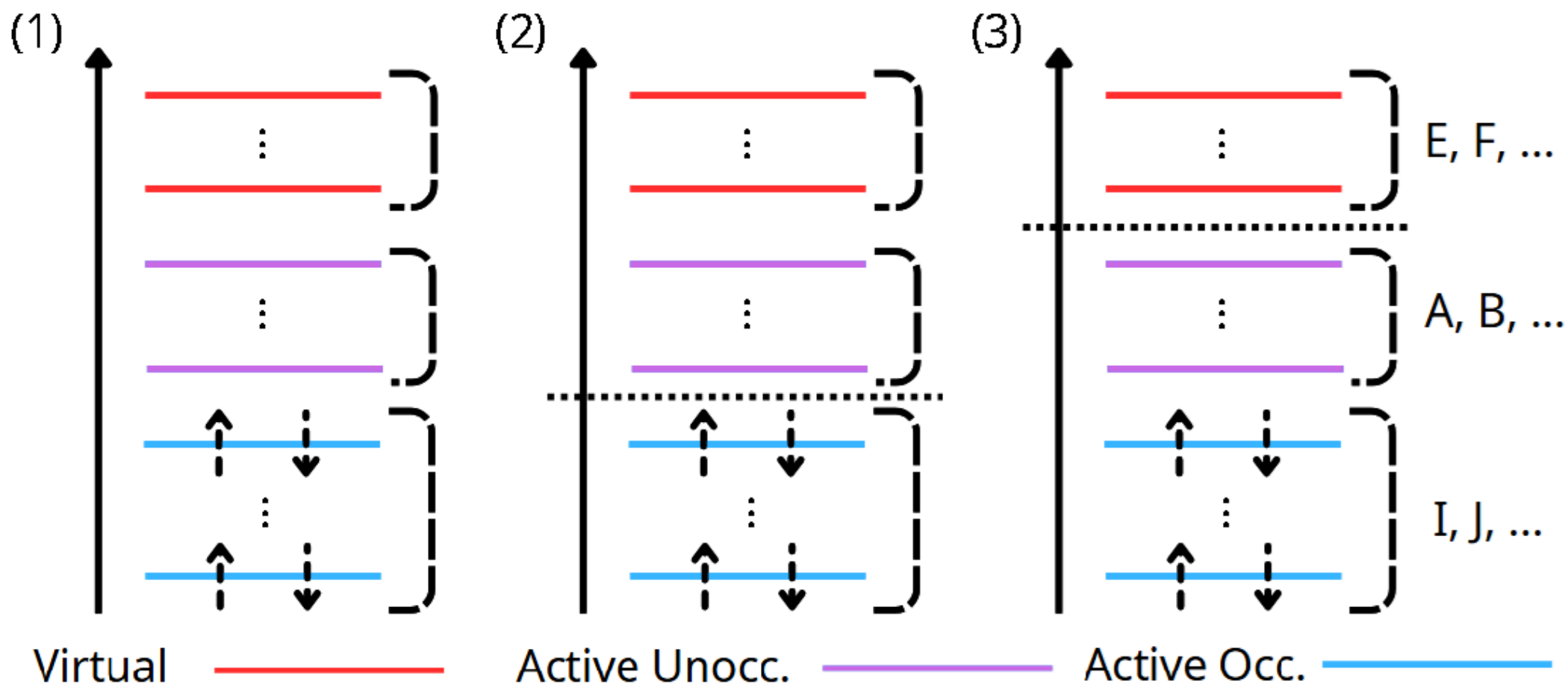
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Theory: Partitioning

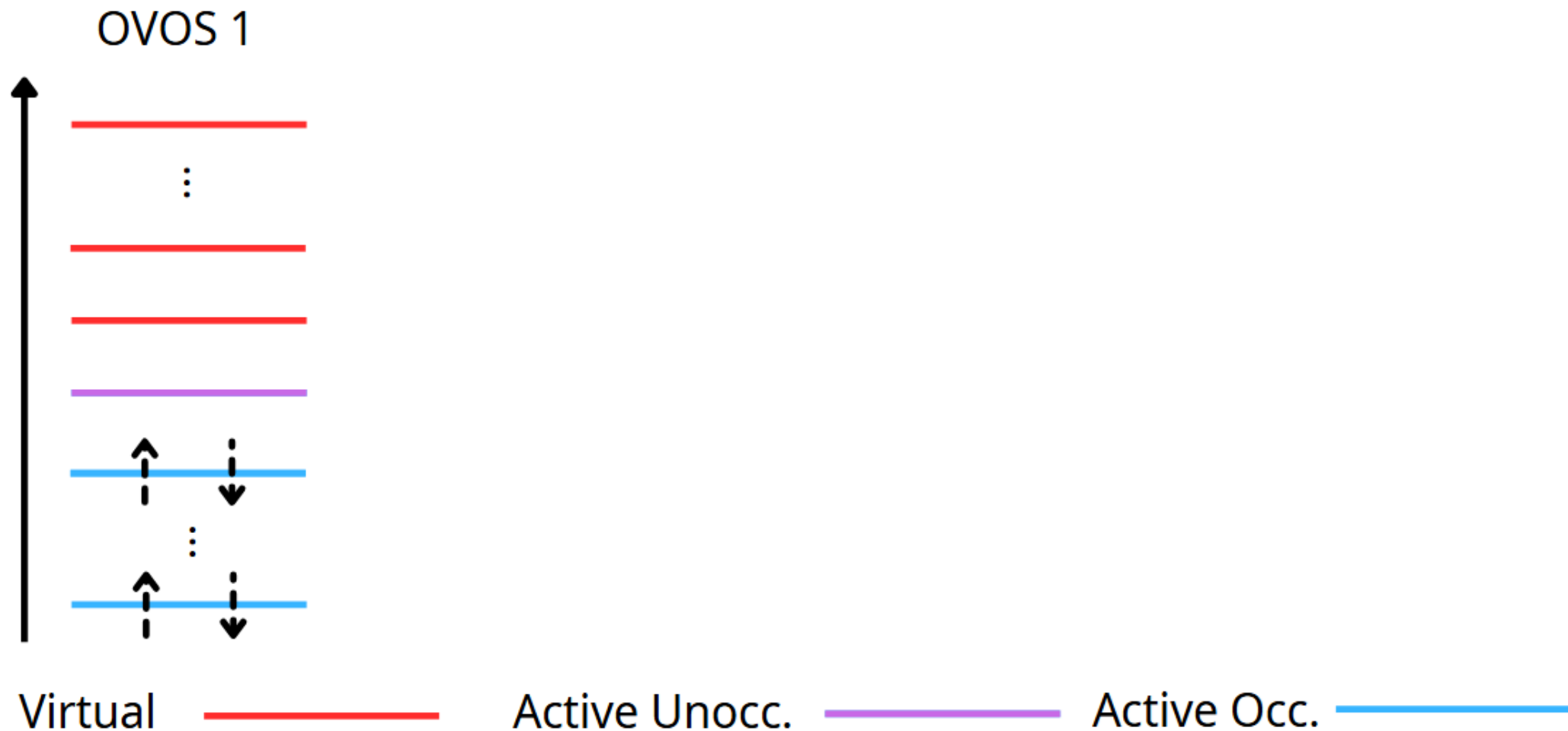
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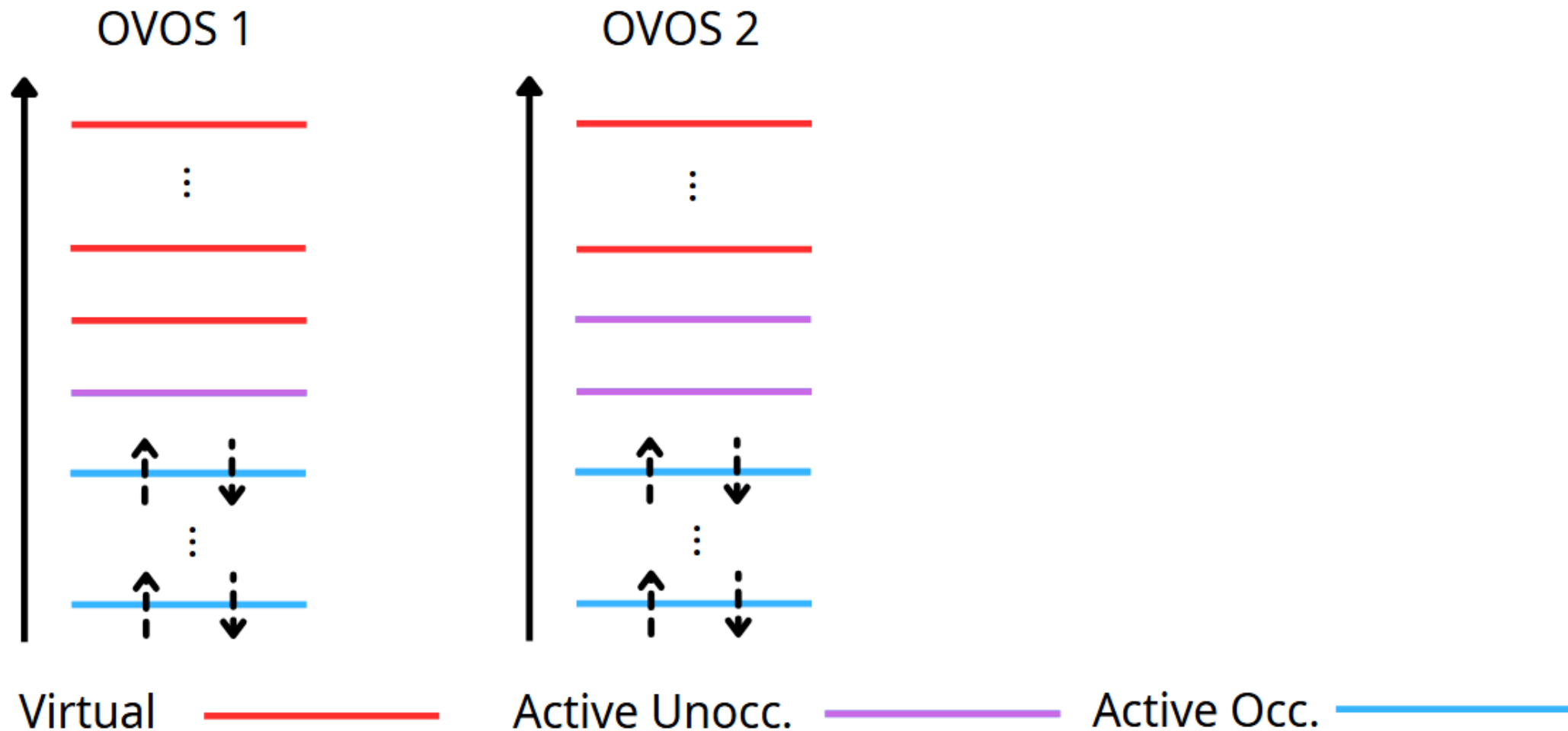
- **Goal:** Find the optimal virtual subspace for capturing electron correlation.



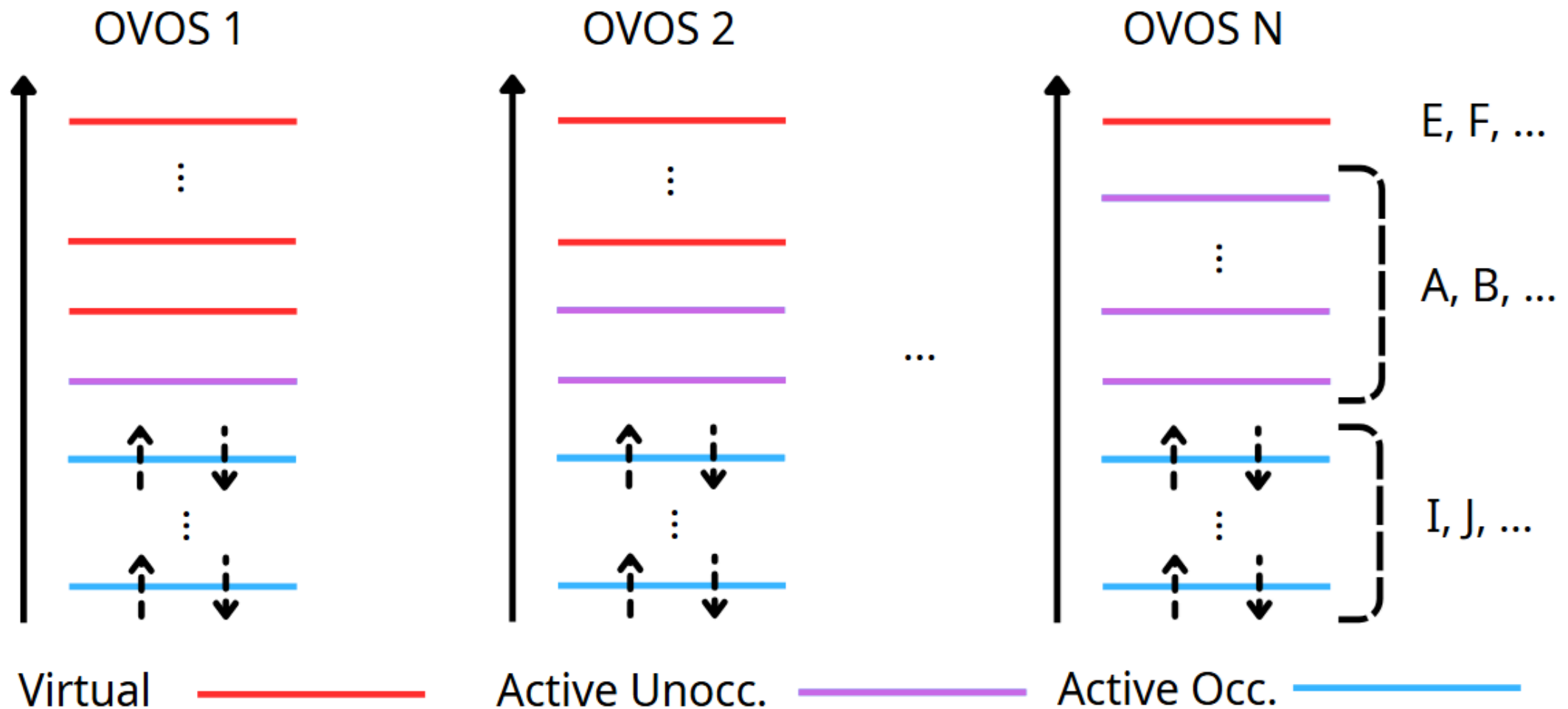
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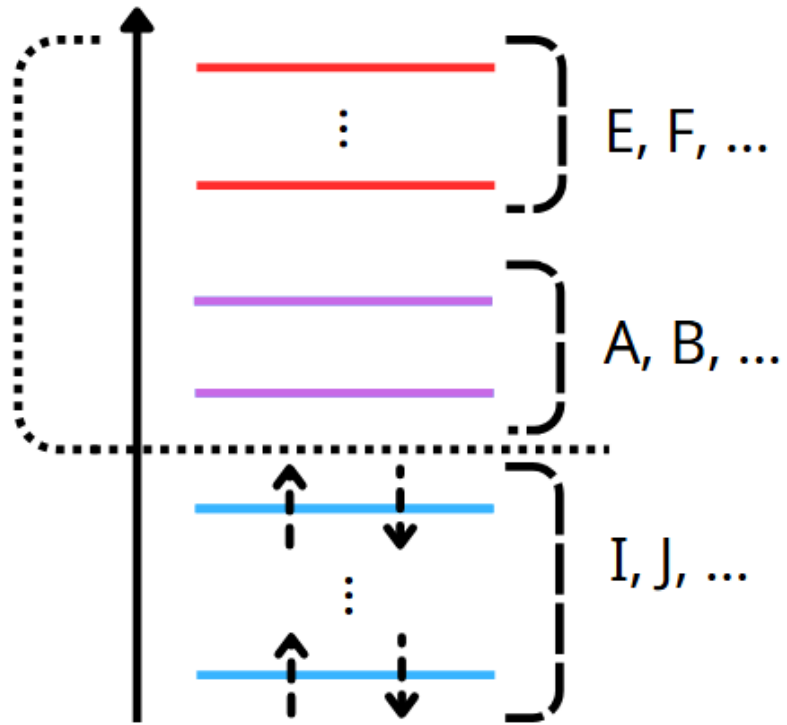


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Theory: Optimisation

(1) Optimize Space



Virtual



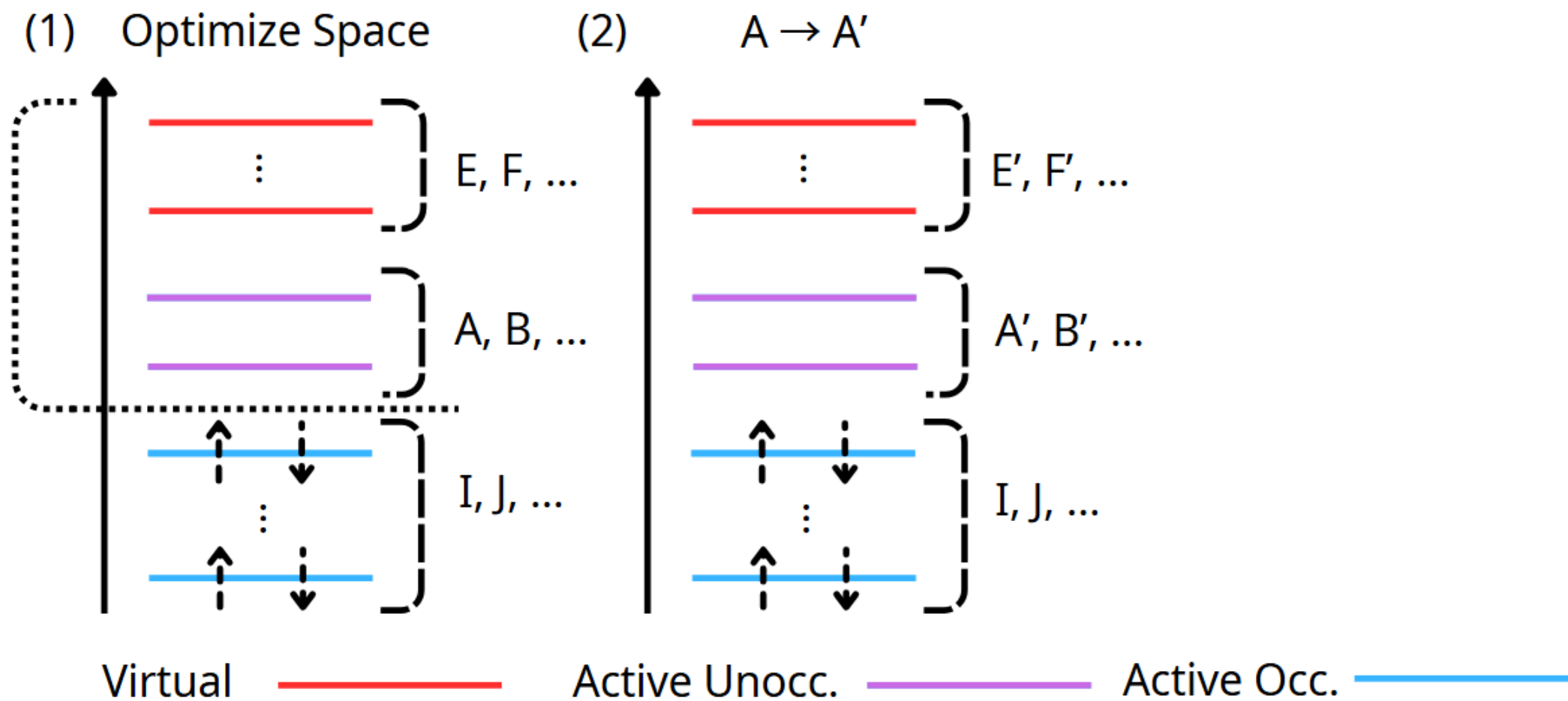
Active Unocc.



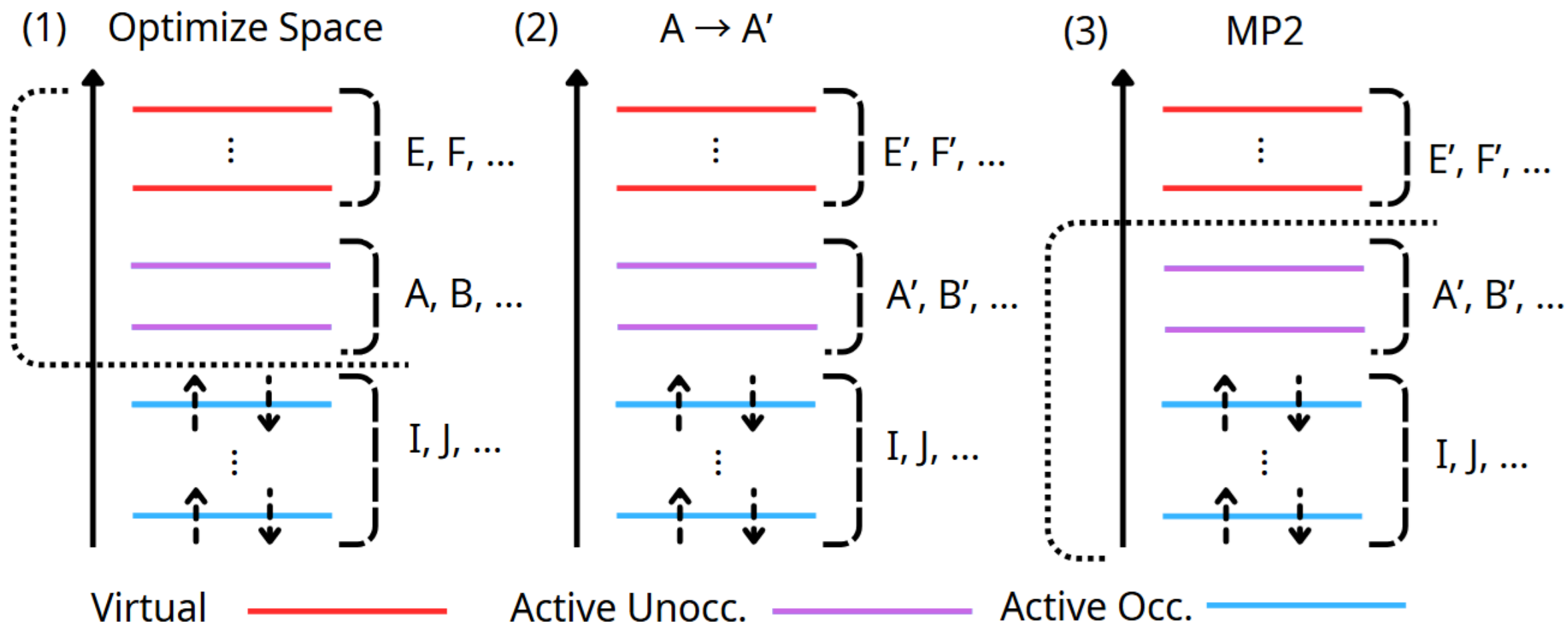
Active Occ.



Theory: Optimisation



Theory: Optimisation



The Hylleraas Functional

- **Framework:** The **Hylleraas functional** is the mathematical target for second-order correlation energy:

$$E^{(2)} \leq \langle \Psi^{(1)} | H_0 - E^{(0)} | \Psi^{(1)} \rangle + 2 \langle \Psi^{(1)} | V - E^{(1)} | \Psi^{(0)} \rangle = J_2$$

- **Functional Form:** Each unique pairs of occupied orbital indices (i, j) defines a functional $J_{ij}^{(2)}$ that depends on the virtual space through the amplitudes t_{ij}^{ab} :

$$J_{ij}^{(2)} = \sum_{a>b, c>d} t_{ij}^{ab} t_{ij}^{cd} (f_{ac} \delta_{bd} - f_{ad} \delta_{bc} + f_{bd} \delta_{ac} - f_{bc} \delta_{ad} - (\epsilon_i + \epsilon_j)(\delta_{ac} \delta_{bd} - \delta_{ad} \delta_{bc})) + 2 \sum_{a>b} t_{ij}^{ab} \langle ij || ab \rangle$$

Derivatives

- **Gradient:** The gradient of $J_{ij}^{(2)}$ with respect to the virtual orbital rotation parameters κ_{ab} is computed analytically.

$$G_{ae} = \frac{\partial J_2}{\partial R_{ae}} = 2 \sum_{i>j} \sum_b t_{ij}^{ab} \langle ij || eb \rangle + 2 \sum_b D_{ab} f_{eb}$$

- **Hessian:** The Hessian matrix is approximated using the orbital energy differences, enabling efficient optimisation.

$$\begin{aligned} H_{ae,bf} = & 2 \sum_{i>j} t_{ij}^{ab} \langle ij || ef \rangle - \sum_{i>j} \sum_c [t_{ij}^{ac} \langle ij || bc \rangle + t_{ij}^{bc} \langle ij || ca \rangle] \delta_{ef} \\ & + D_{ab} (f_{aa} - f_{bb}) \delta_{ef} + D_{ab} f_{ef} (1 - \delta_{ef}) \end{aligned}$$

Newton–Raphson

Rotation: We use the Newton–Raphson method to solve for the optimal rotation parameters R that minimize the Hylleraas functional.

$$R = -G \cdot H^{-1} \quad \longrightarrow \quad H \cdot R = -G$$

Unitary Rotation

Coefficients: We perform a unitary rotation of the virtual orbitals using the computed rotation parameters R to generate the new orbital coefficients.

$$U = e^R \quad \longrightarrow \quad UCU^\dagger = C' \quad (U^\dagger U = I)$$

Canonicalization

Orbital Canonicalization: After a rotation, we canonicalize the active unoccupied Fock block to maintain the orbital energies and ensure stability of the optimisation.

$$F_{ab}^{\text{active}} = C_{ab}'^{\dagger} F C_{ab}' \longrightarrow \text{Diag}(F_{ab}^{\text{active}}) = C''$$

Method

Implementation - Computational Details



Method: Implementation

Software: Implemented and benchmarked OVOS w. PySCF, and SlowQuant.

Algorithm 1: OVOS Iterative Optimisation

- 1 **Perform** SCF calculation
- 2 **Transform** AO integrals to MO basis
- 3 **Select** Initial virtual orbital space
- 4 **Repeat**
 - 5 Compute MP1 amplitudes and MP2
 - 6 Assemble gradient and Hessian
 - 7 Solve Newton–Raphson
 - 8 Generate Unitary Matrix
 - 9 Update MO coefficients
- 10 Canonicalize active Fock block
- 11 Evaluate new MP2
- 12 **Until** convergence

Perform SCF calculation to obtain canonical orbitals.

```
# Water molecule, minimal basis
mol = pyscf.gto.Mole()
mol.atom = 'C 0 0 0; 0 0 0 1.128'
mol.basis = '6-31G'
mol.unit = 'Angstrom'
mol.spin = 0
mol.charge = 0
mol.symmetry = False
mol.verbose = 0
mol.build()

# RHF reference
mf = pyscf.scf.RHF(mol)
mf.verbose = 0
mf.kernel()

# Initial data (RHF orbitals)
Fao = [mf.get_fock(), mf.get_fock()]
mo_coeffs = [mf.mo_coeff, mf.mo_coeff]

# Create OVOS object and run
ovos = OVOS(mol=mol,
            scf=mf,
            Fao=Fao,
            num_opt_virtual_orbs=2,
            mo_coeff=mo_coeffs,
            init_orbs="RHF",
            verbose=1,
            max_iter=1000,
            conv_energy=1e-8,
            conv_grad=1e-6,
```

Transform AO integrals to MO basis.

```
# AO integrals
self.S = mol.intor('int1e_ovlp')
self.hcore_ao = mol.intor("int1e_kin") + mol.intor("int1e_nuc")
self.eri_4fold_ao = mol.intor('int2e_sph', aosym=1)

...

# -----
# Integral transformation
# -----
def _eri_vovo_antisym(self, mo_coeffs: List[np.ndarray]) → np.ndarray:
    """
    Transform (vir,occ|vir,occ) AO integrals to MO basis, assemble spin-orbital
    and antisymmetrize. Returns <ab||ij> in physicist notation (a,b,i,j).
    """
    nocc_a, nocc_b = self.mol.nelec
    C_occ_a = mo_coeffs[0][:, :nocc_a]
    C_occ_b = mo_coeffs[1][:, :nocc_b]
    C_vir_a = mo_coeffs[0][:, nocc_a:]
    C_vir_b = mo_coeffs[1][:, nocc_b:]
    nvir_a = C_vir_a.shape[1]
    nvir_b = C_vir_b.shape[1]

    def _ao2mo(C_left, C_row, C_right, C_col):
        # returns (left,row|right,col) in chemist notation
        return pyscf.ao2mo.kernel(self.eri_4fold_ao,
                                   [C_left, C_row, C_right, C_col],
                                   compact=False)

    # Chemist blocks (vir,occ|vir,occ)
    vovo_aaaa = _ao2mo(C_vir_a, C_occ_a, C_vir_a, C_occ_a).reshape(
        nvir_a, nocc_a, nvir_a, nocc_a)
    vovo_bbbb = _ao2mo(C_vir_b, C_occ_b, C_vir_b, C_occ_b).reshape(
        nvir_b, nocc_b, nvir_b, nocc_b)
    vovo_aabb = _ao2mo(C_vir_a, C_occ_a, C_vir_b, C_occ_b).reshape(
        nvir_a, nocc_a, nvir_b, nocc_b)

    nvir_spin = 2 * nvir_a
    nocc_spin = nocc_a + nocc_b

    # Assemble spin-orbital chemist tensor
    vovo_spin = np.zeros((nvir_spin, nocc_spin, nvir_spin, nocc_spin))
    vovo_spin[0::2, 0::2, 0::2, 0::2] = vovo_aaaa
    vovo_spin[1::2, 1::2, 1::2, 1::2] = vovo_bbbb
    vovo_spin[0::2, 0::2, 1::2, 1::2] = vovo_aabb
    vovo_spin[1::2, 1::2, 0::2, 0::2] = vovo_aabb.transpose(2, 3, 0, 1)
```

Select initial virtual orbital space (canonical HF, previous OVOS, or random).

```
mf = scf.RHF(mol)
```

Repeat until convergence: ...

```
mf = scf.RHF(mol)
```



Results

OVOS - VQE - ooVQE



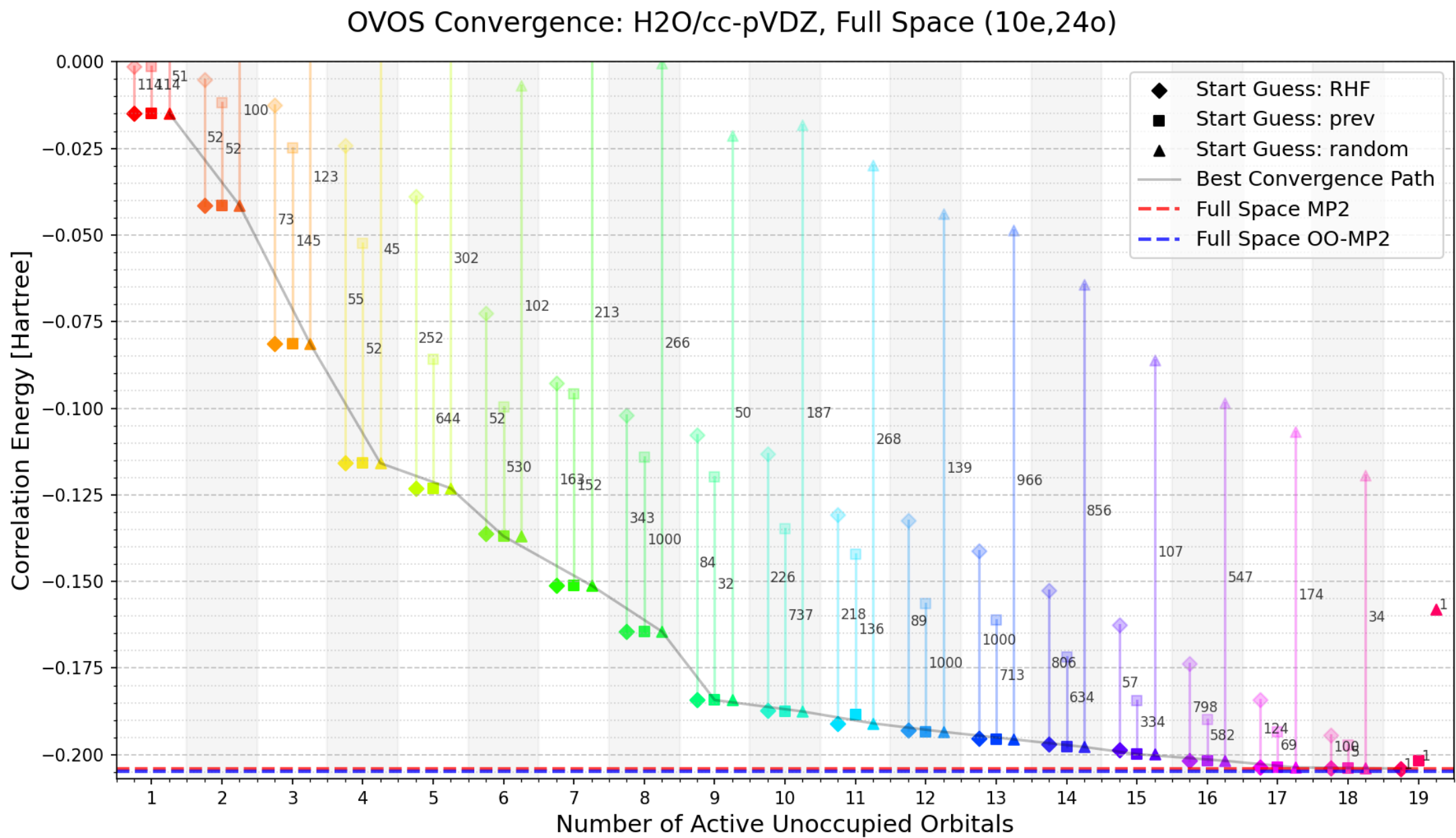
Computational Details

- **Molecules:** H₂O, CO, HF, NH₃, Li₂.
- **Basis Set:** 6-31G, cc-pVDZ.
- **Start guesses:** Canonical HF virtual orbitals, previous OVOS, and random.
- **Convergence Criteria:** Energy change < 1e-8 a.u., gradient norm < 1e-6.
- **Software:** PySCF for SCF and integral transformations. SlowQuant for unrestricted wavefunction in VQE.

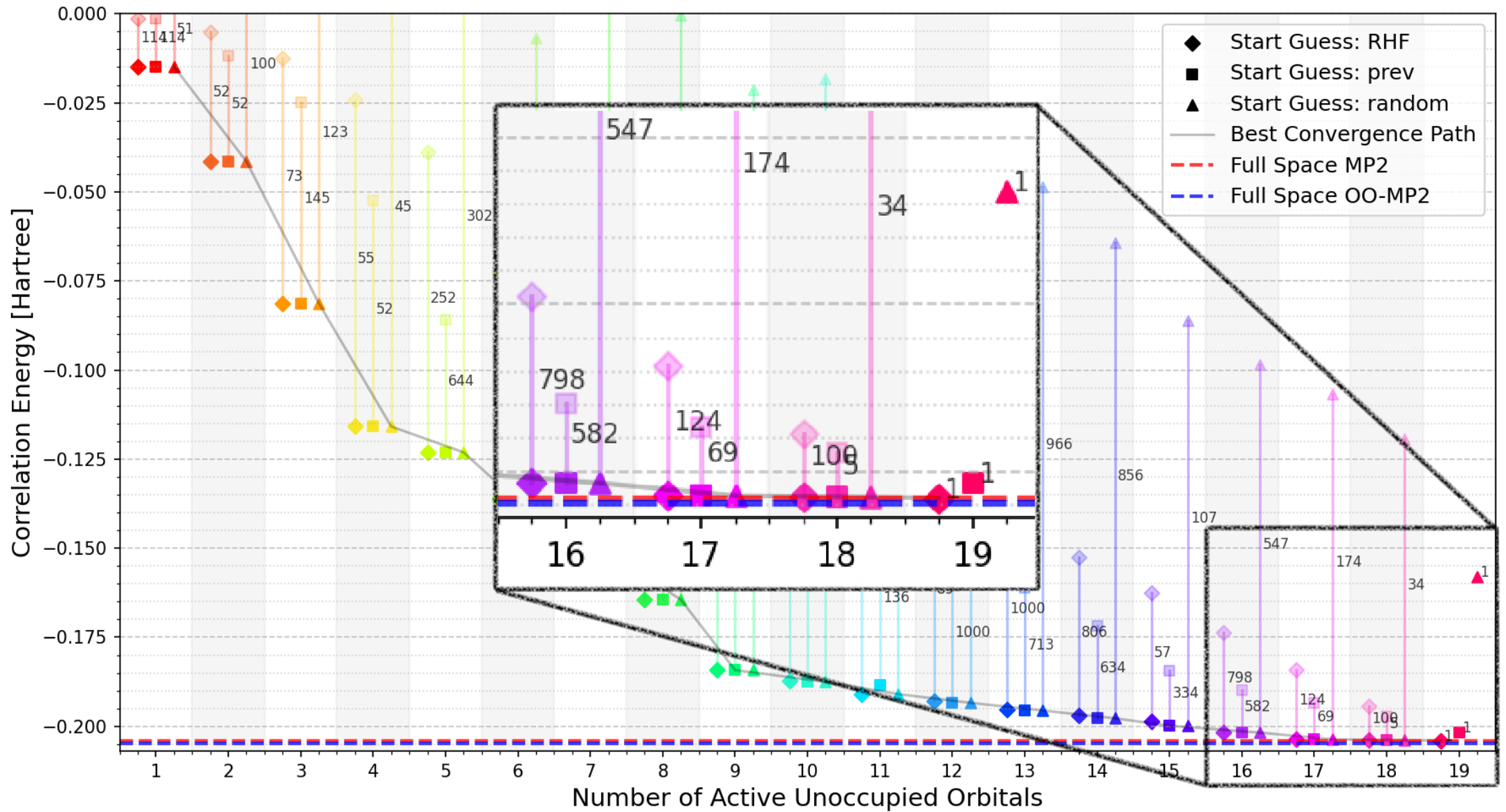
Results: OVOS

OVOS plots

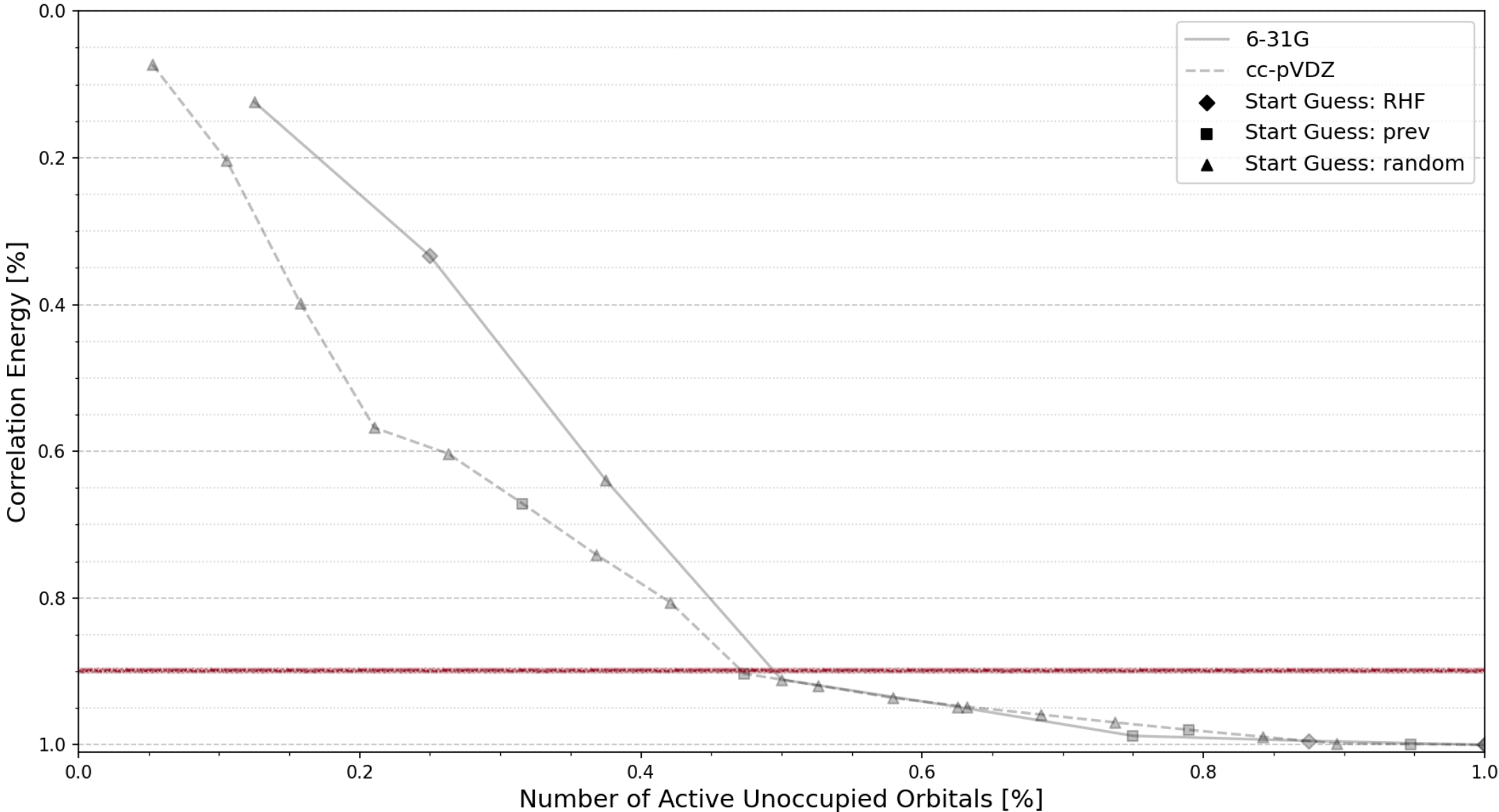
- *Points*: MP2 correlation energy at each iteration.
- *Lines*: Amount of recovered correlation energy from initial point to final OVOS solution.
- *Grey Line*: Best of start Guesses (canonical HF, previous OVOS, random).
- *Dashed Lines*: Reference MP2, and orbital optimised MP2 (both with full virtual space).



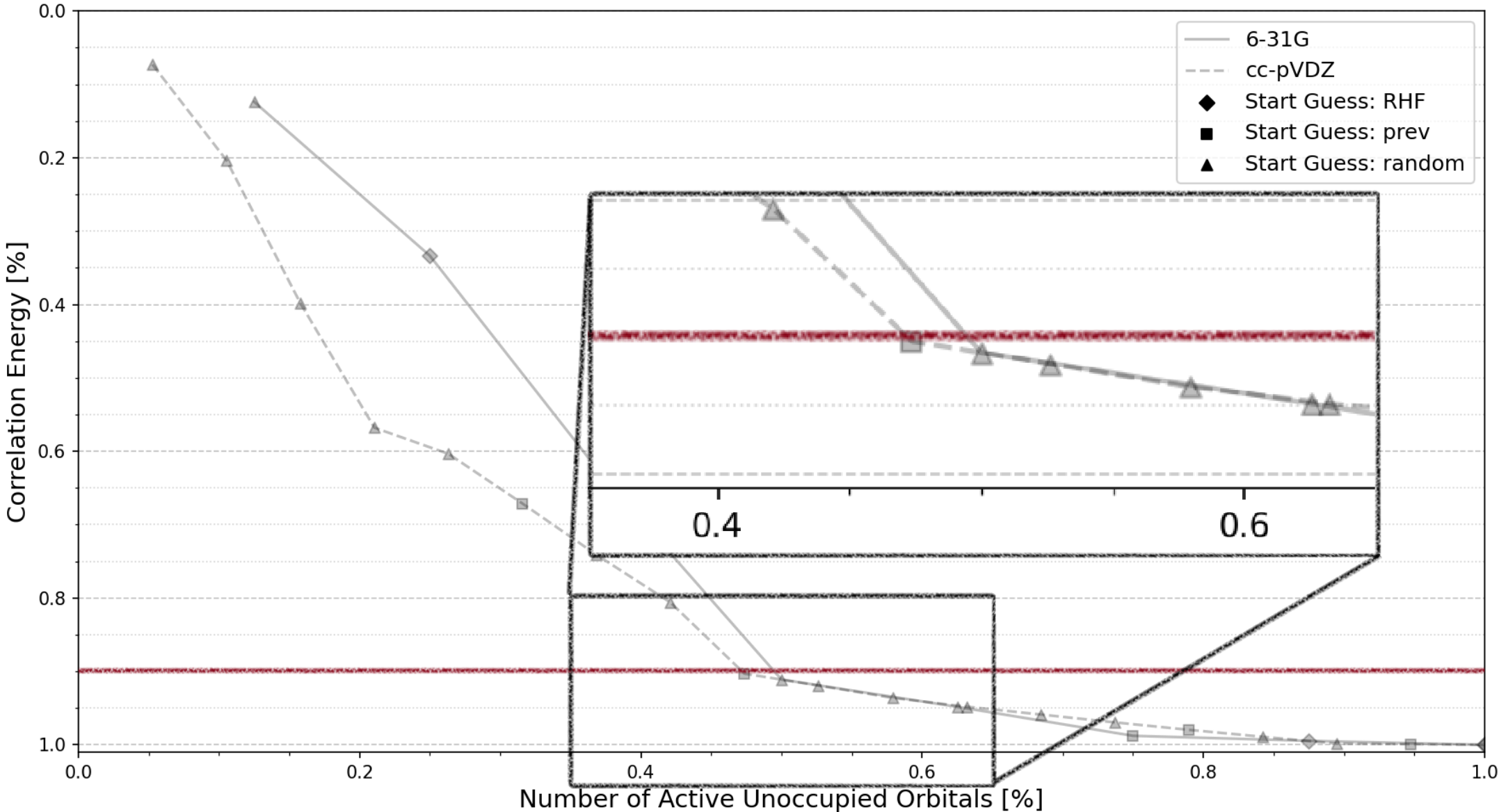
OVOS Convergence: H2O/cc-pVDZ, Full Space (10e,24o)

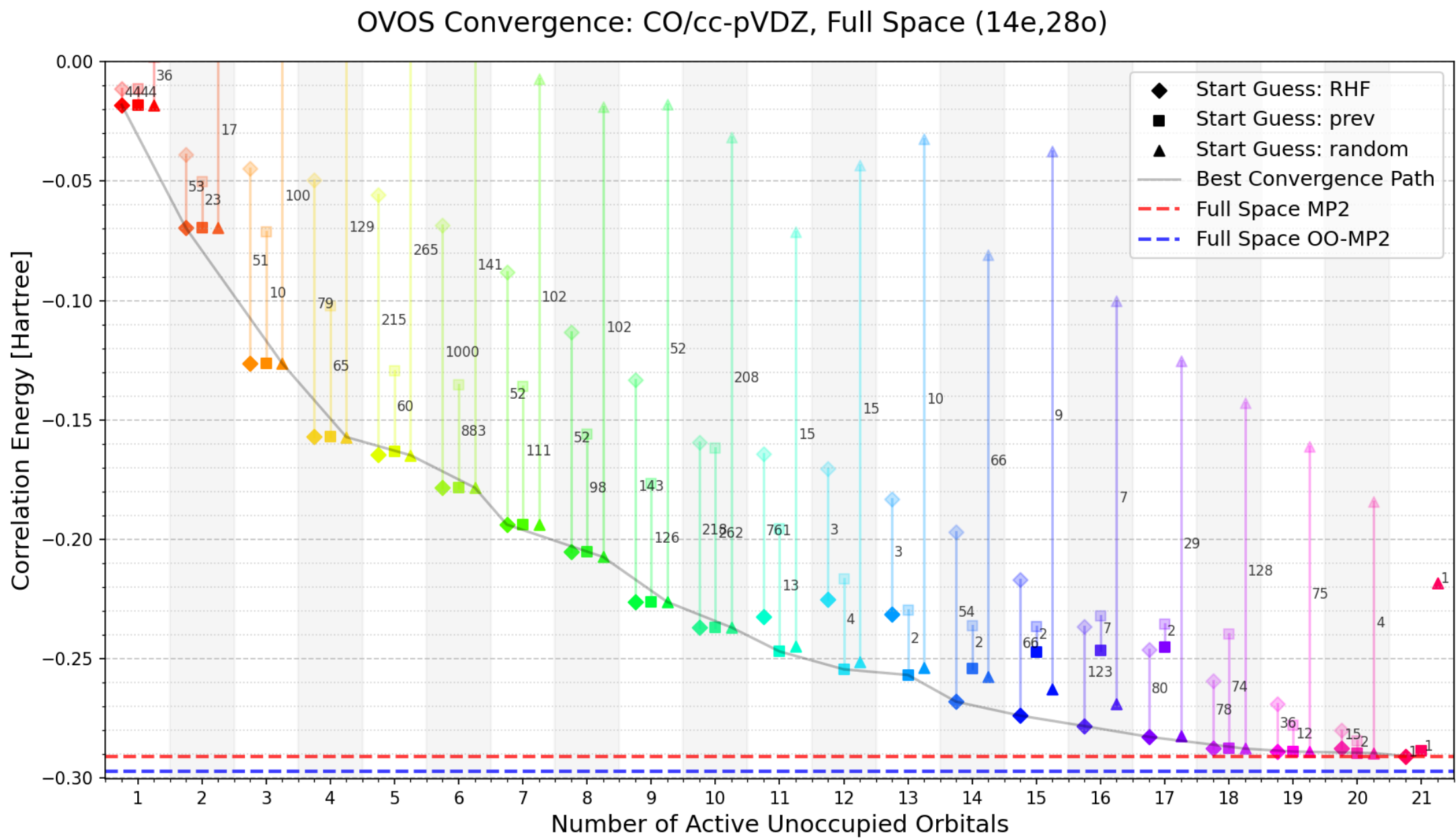


OVOS Basis set convergence: H2O/(6-31G, cc-pVDZ), Full Space (10e,(13, 24)o)

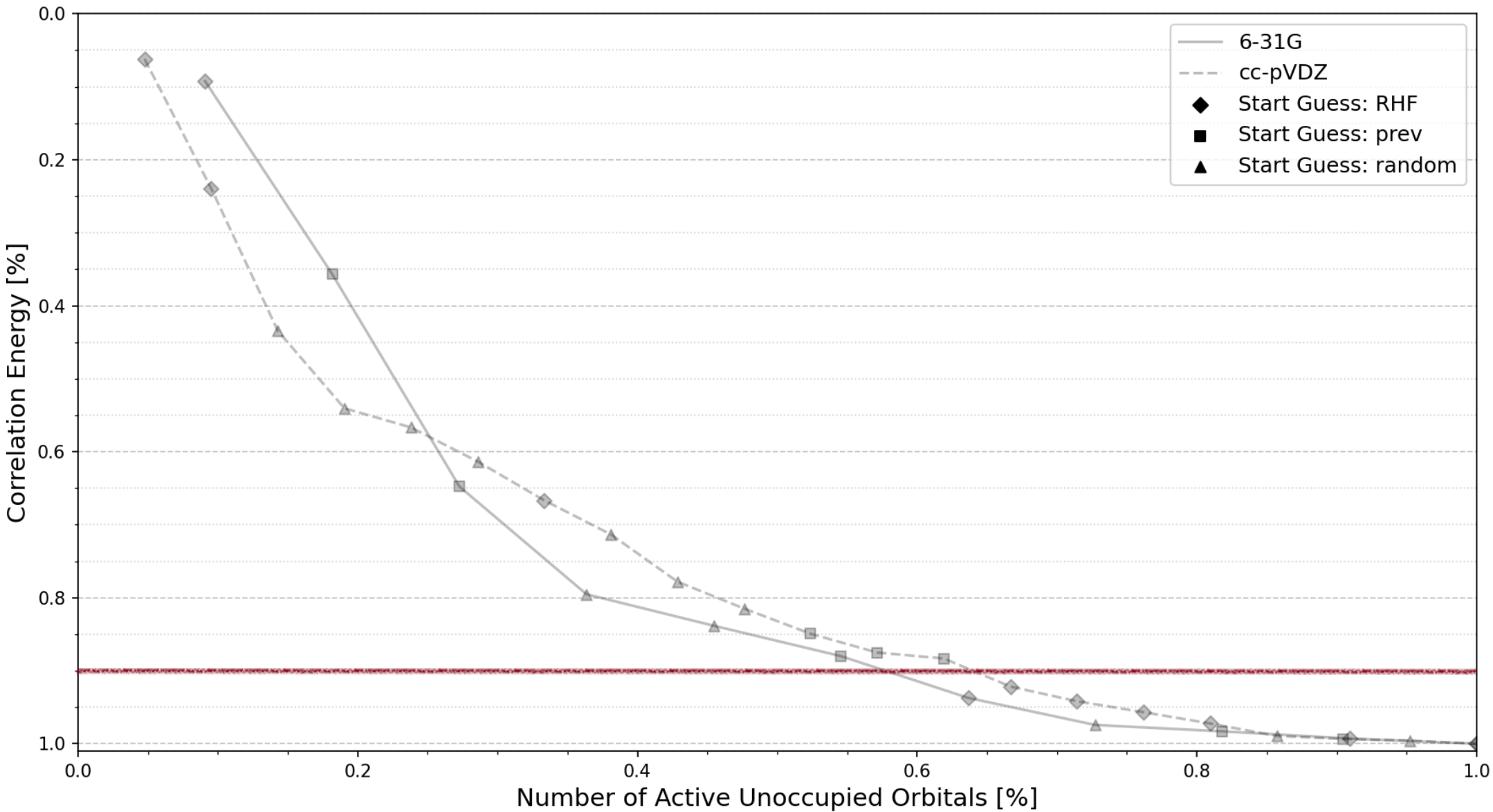


OVOS Basis set convergence: H2O/(6-31G, cc-pVDZ), Full Space (10e,(13, 24)o)

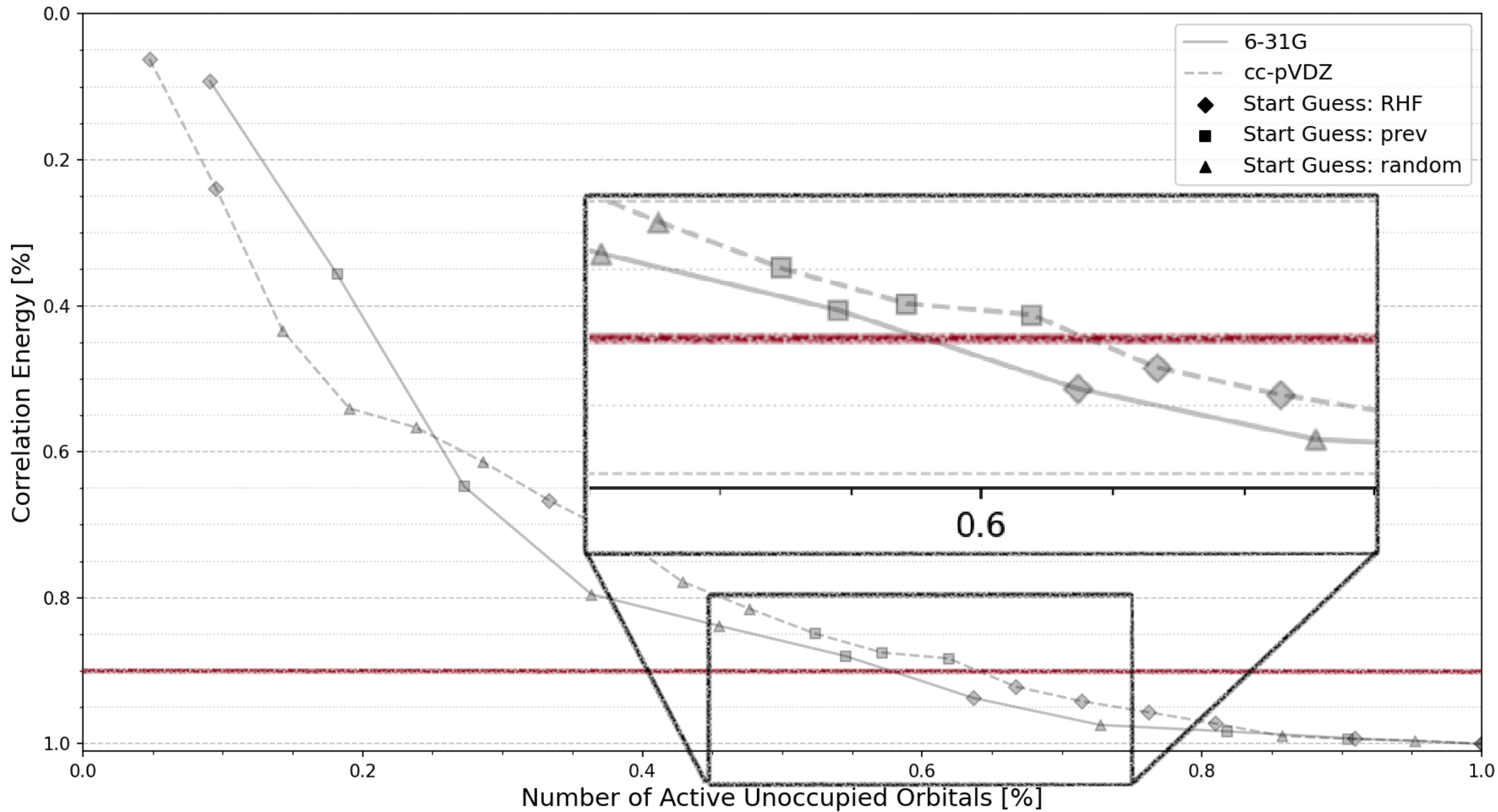




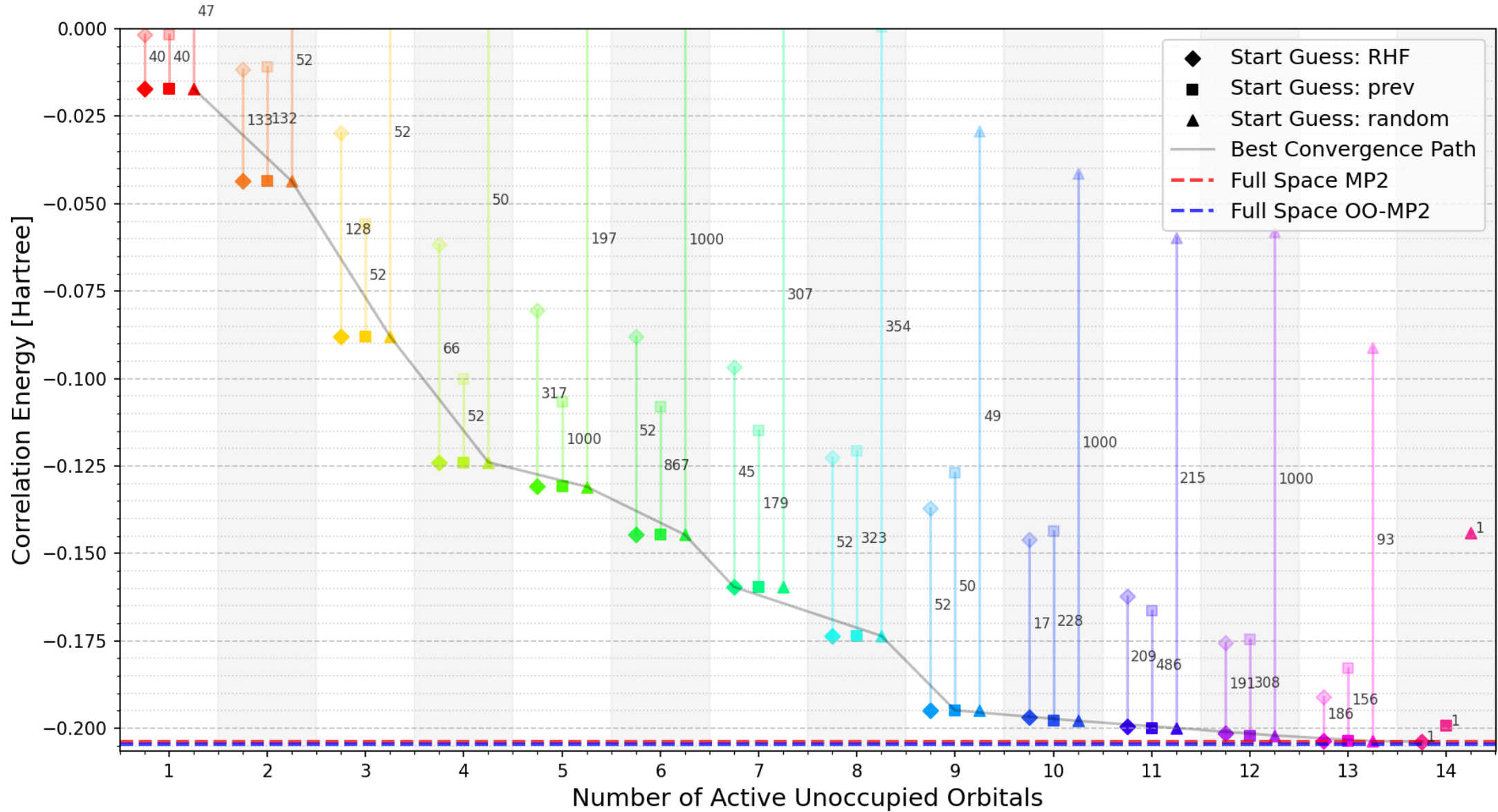
OVOS Basis set convergence: CO/(6-31G, cc-pVDZ), Full Space (14e,(18, 28)o)



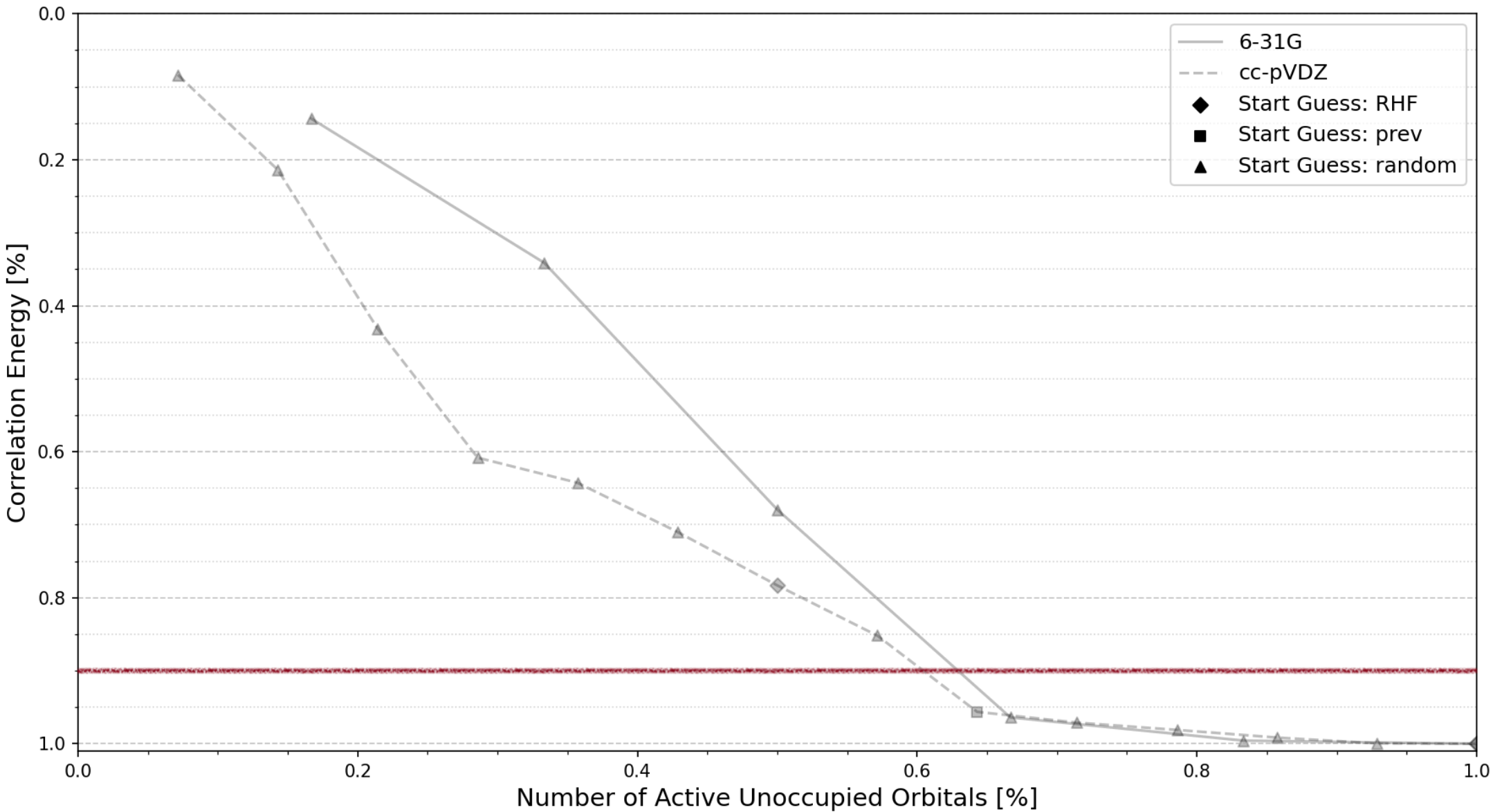
OVOS Basis set convergence: CO/(6-31G, cc-pVDZ), Full Space (14e,(18, 28)o)



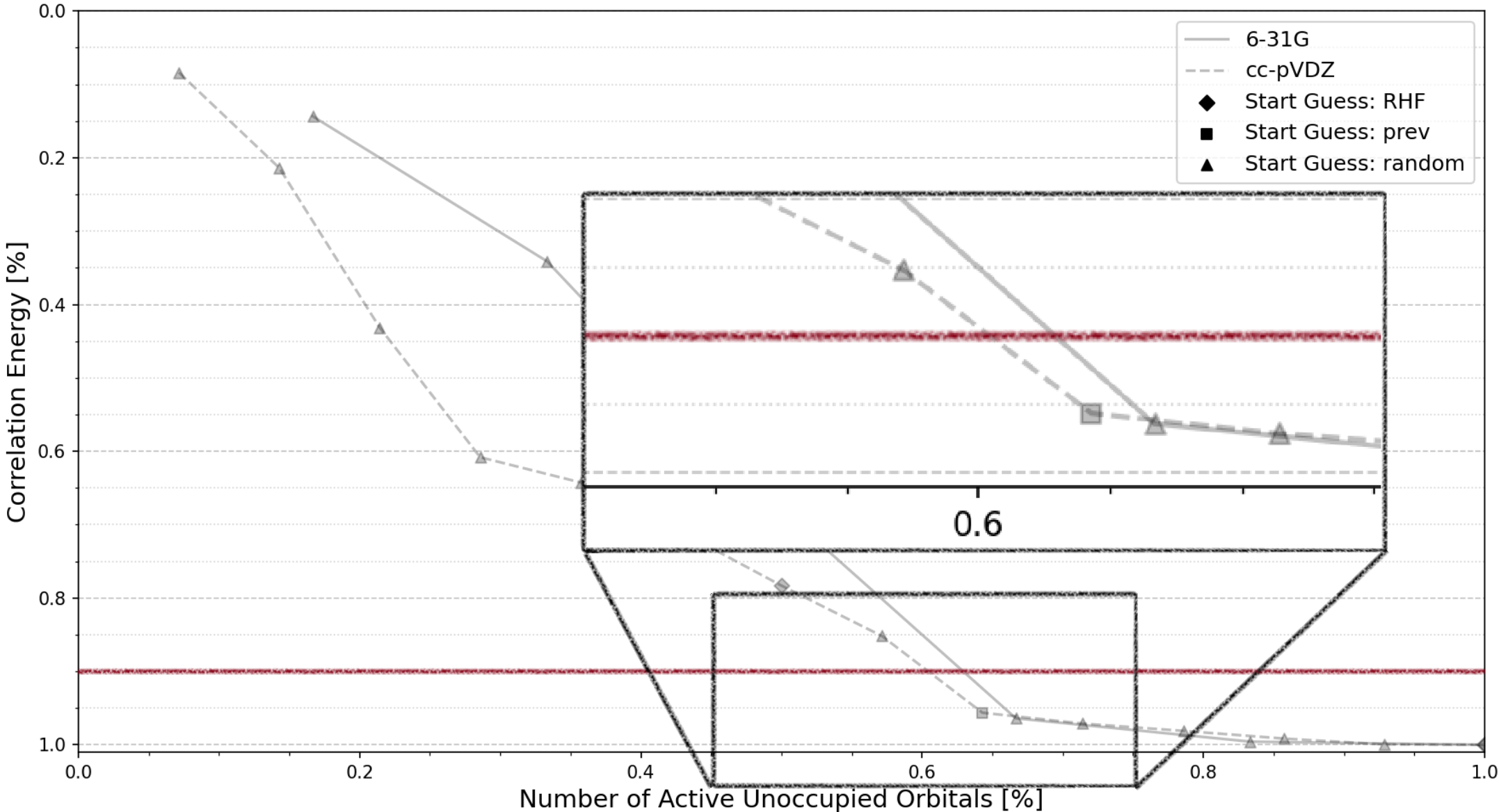
OVOS Convergence: HF/cc-pVDZ, Full Space (10e,19o)



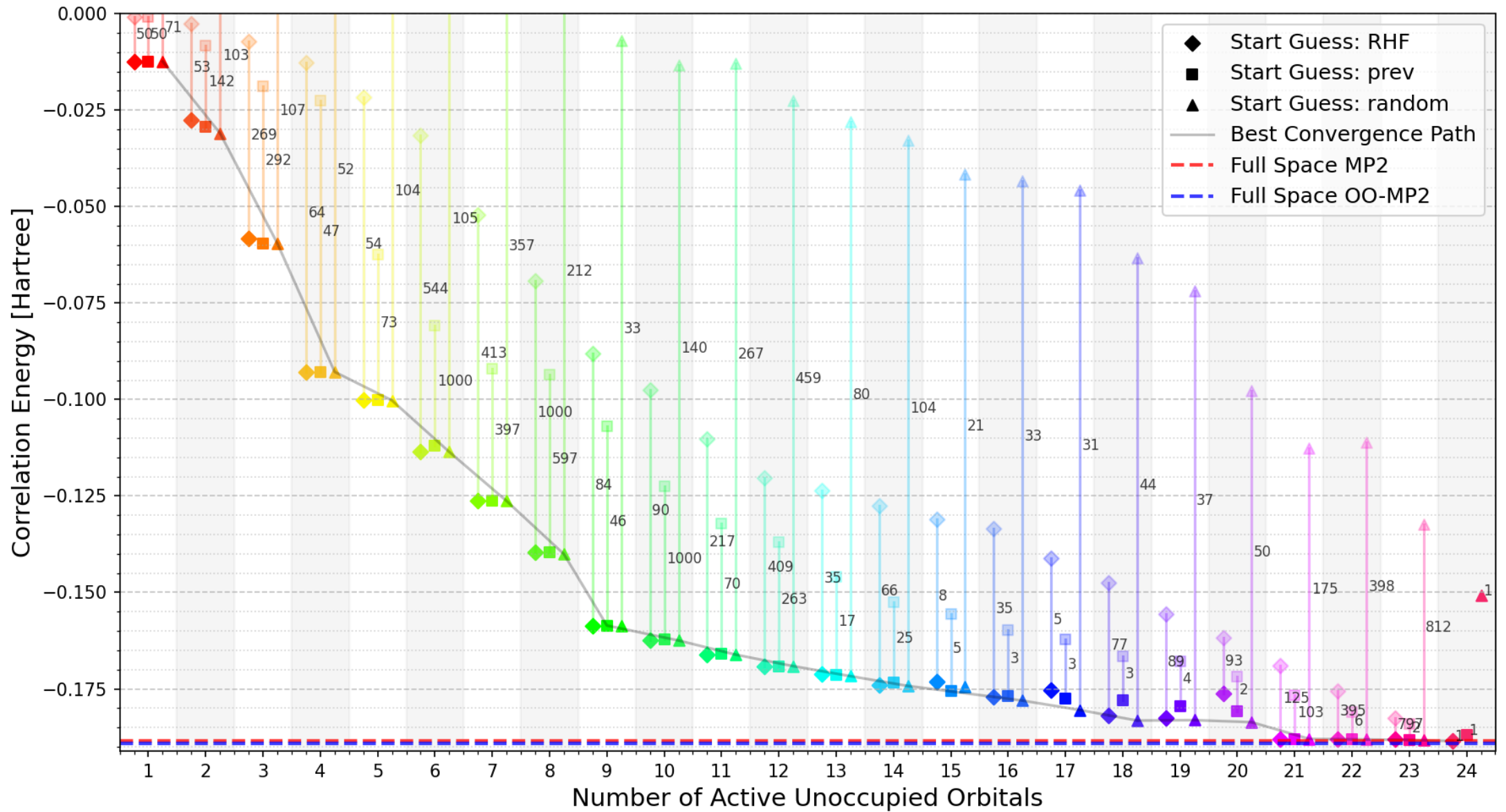
OVOS Basis set convergence: HF/(6-31G, cc-pVDZ), Full Space (10e,(11, 19)o)



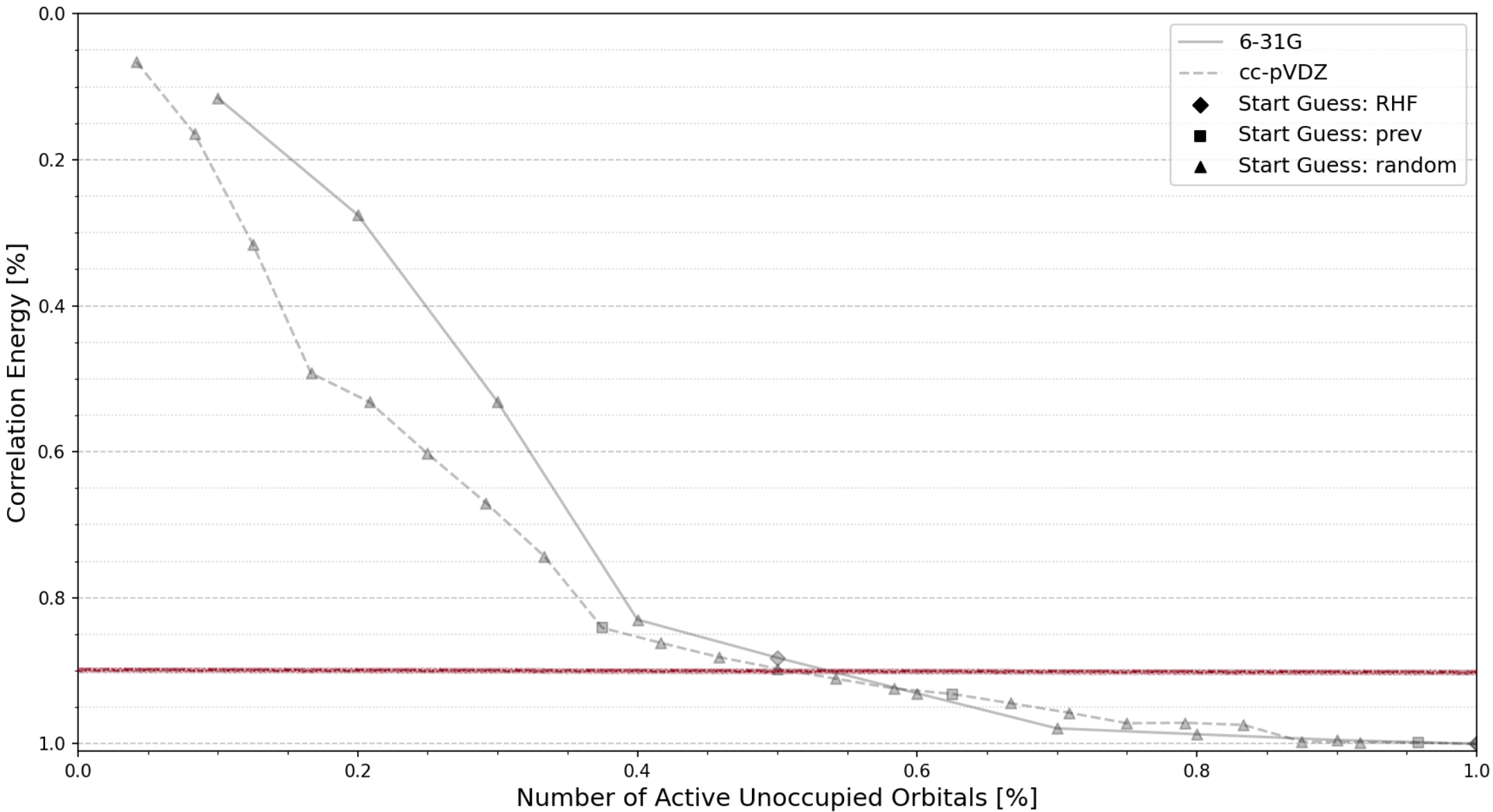
OVOS Basis set convergence: HF/(6-31G, cc-pVDZ), Full Space (10e,(11, 19)o)



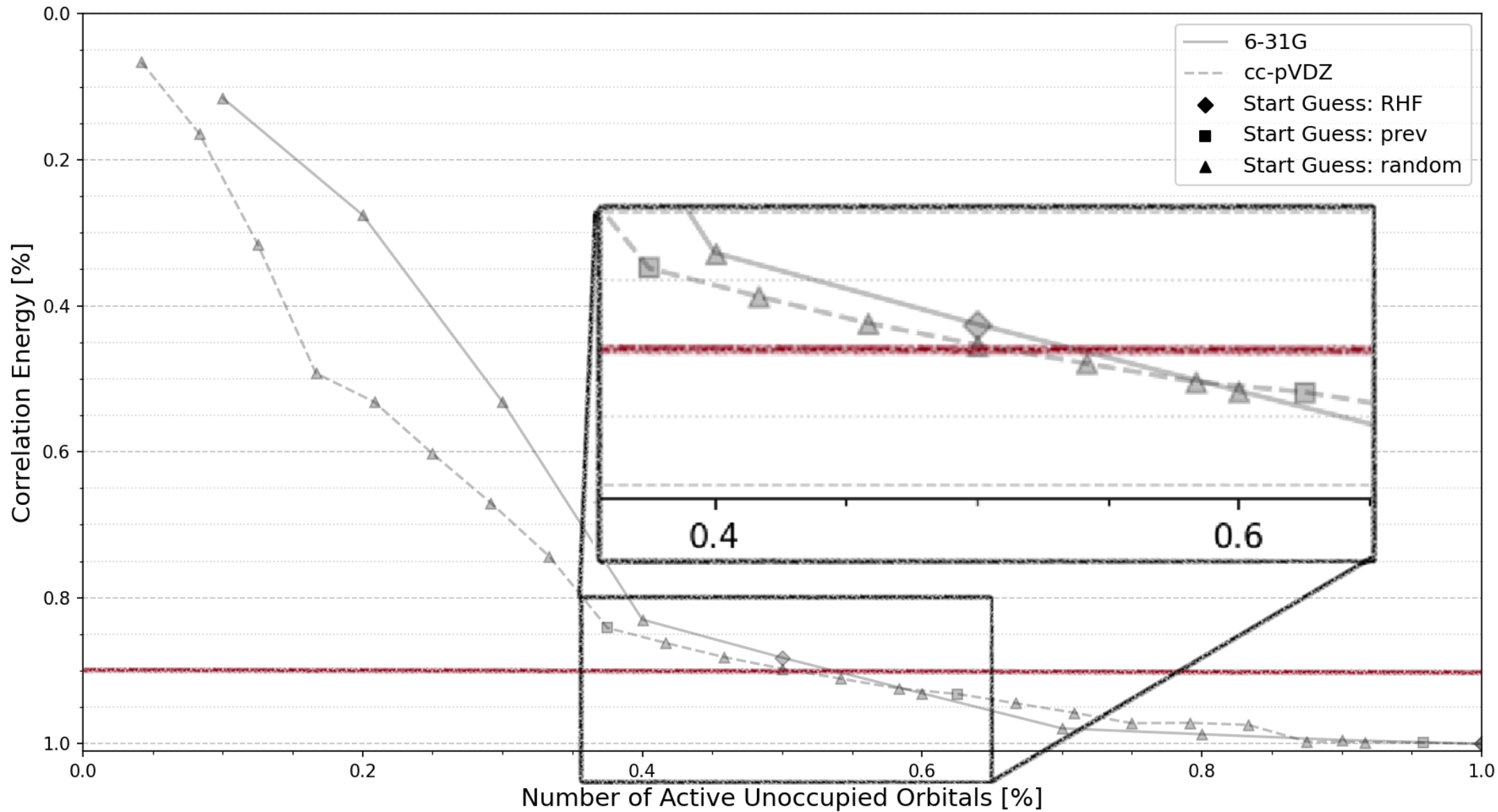
OVOS Convergence: NH3/cc-pVDZ, Full Space (10e,29o)

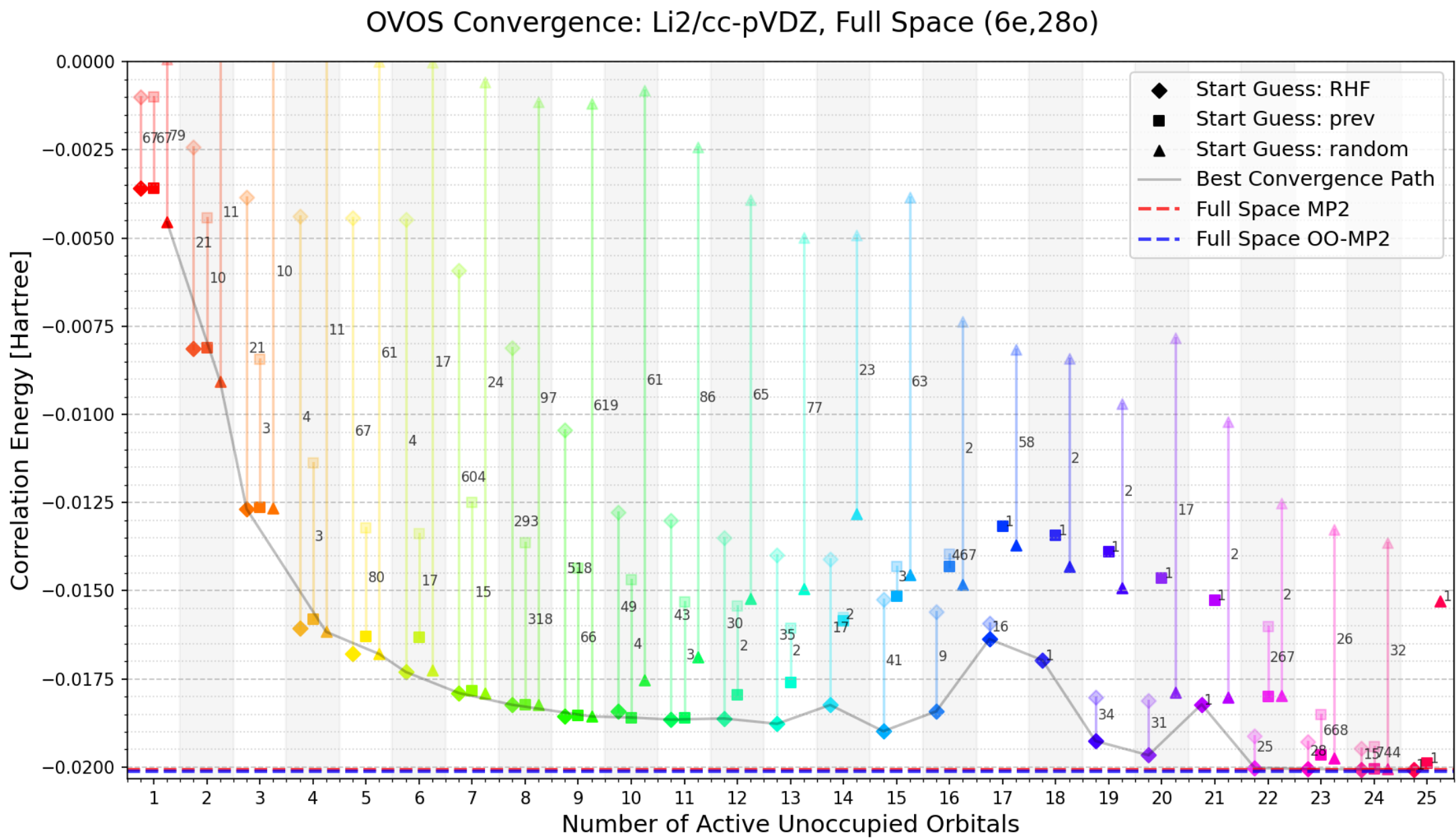


OVOS Basis set convergence: NH3/(6-31G, cc-pVDZ), Full Space (10e,(15, 29)o)

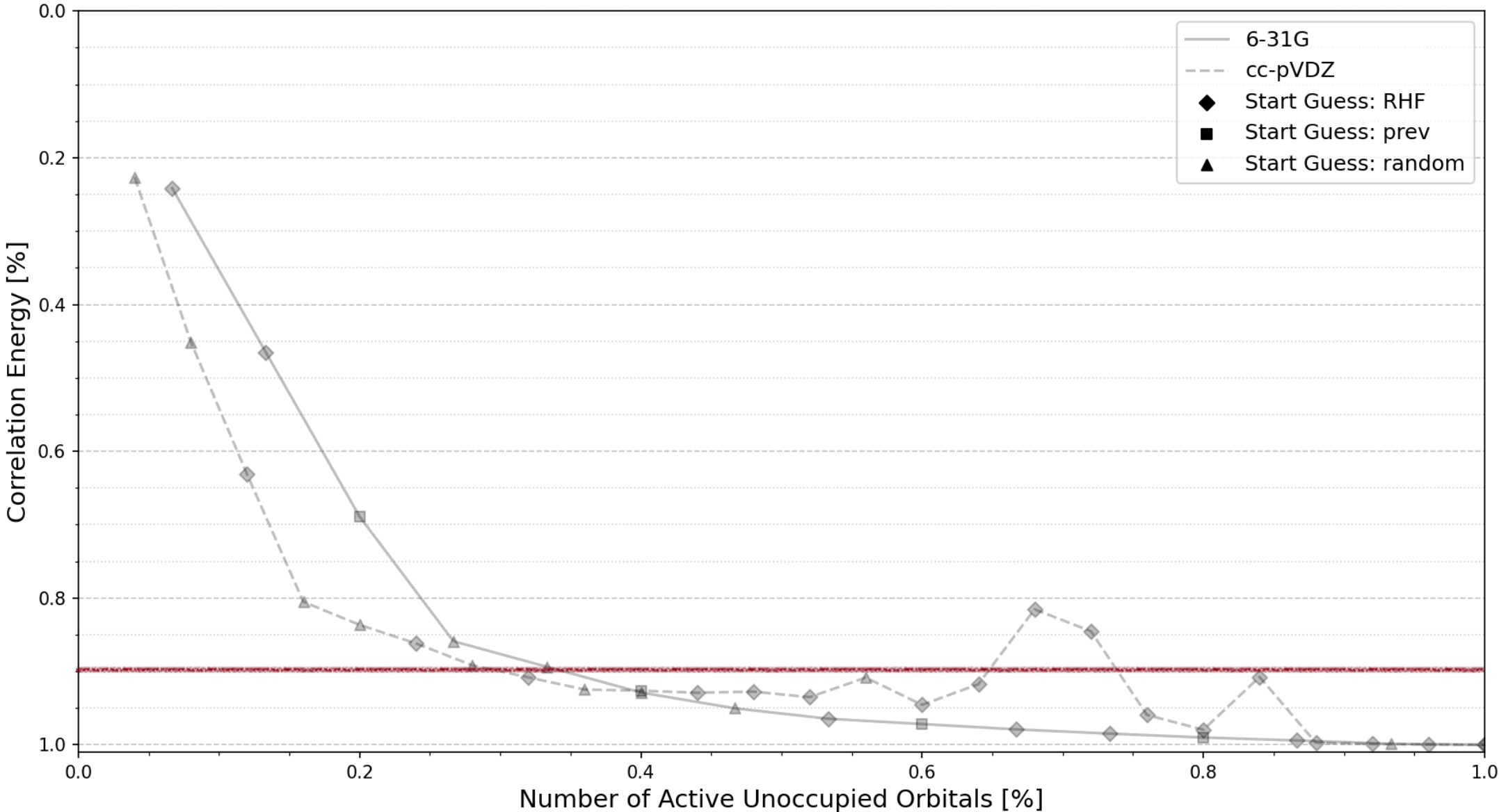


OVOS Basis set convergence: NH3/(6-31G, cc-pVDZ), Full Space (10e,(15, 29)o)

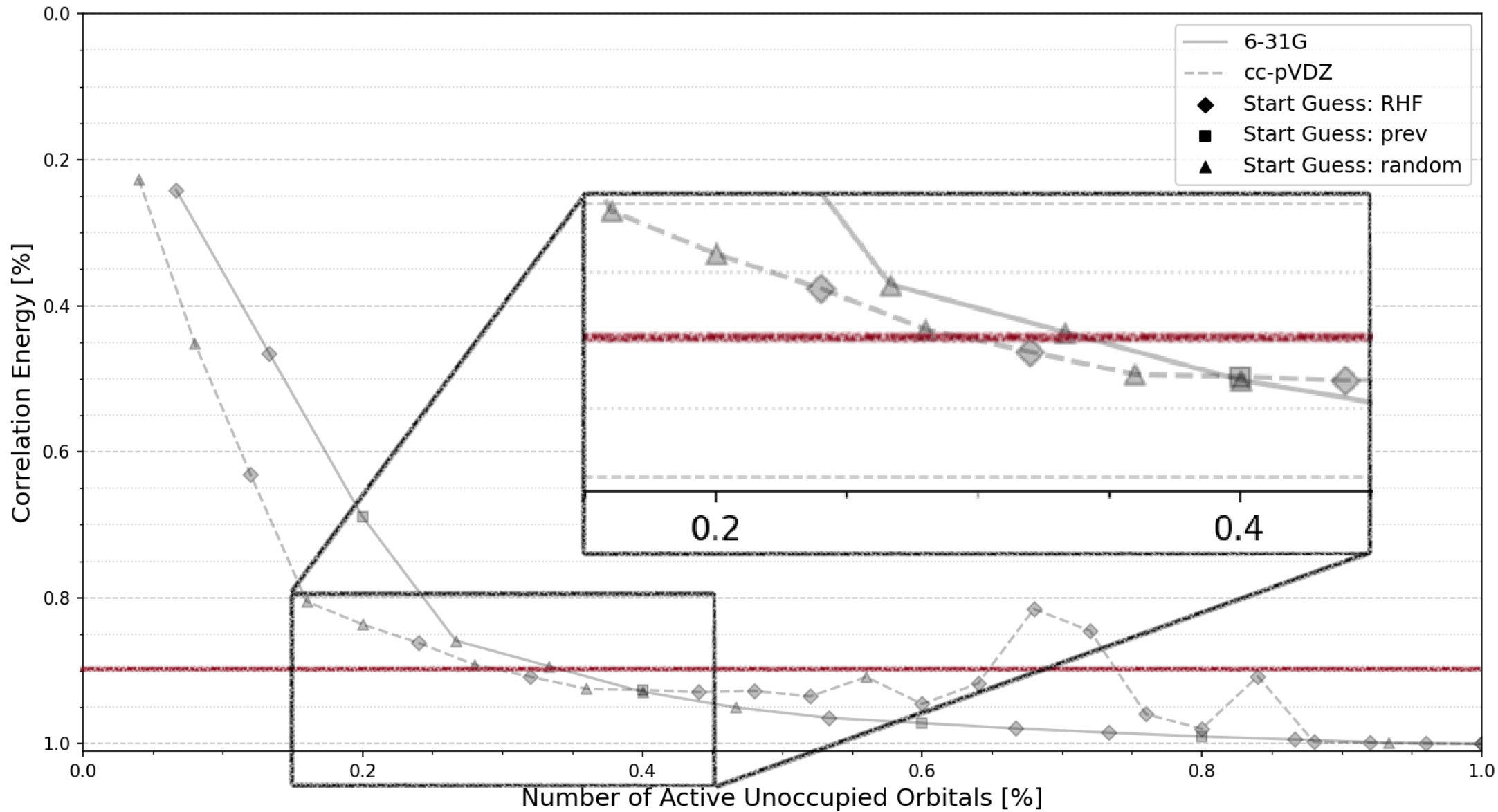




OVOS Basis set convergence: Li2/(6-31G, cc-pVDZ), Full Space (6e,(18, 28)o)



OVOS Basis set convergence: Li2/(6-31G, cc-pVDZ), Full Space (6e,(18, 28)o)



Results: OVOS Summarized

Molecule	Fullspace*	N'_{virt} (90% MP2)	% of full space
H ₂ O	19	13	68%
CO	21	13	62%
HF	14	9	64%
NH ₃	24	12	50%
Li₂	25	7	28%

*Virtual space size in cc-pVDZ basis.

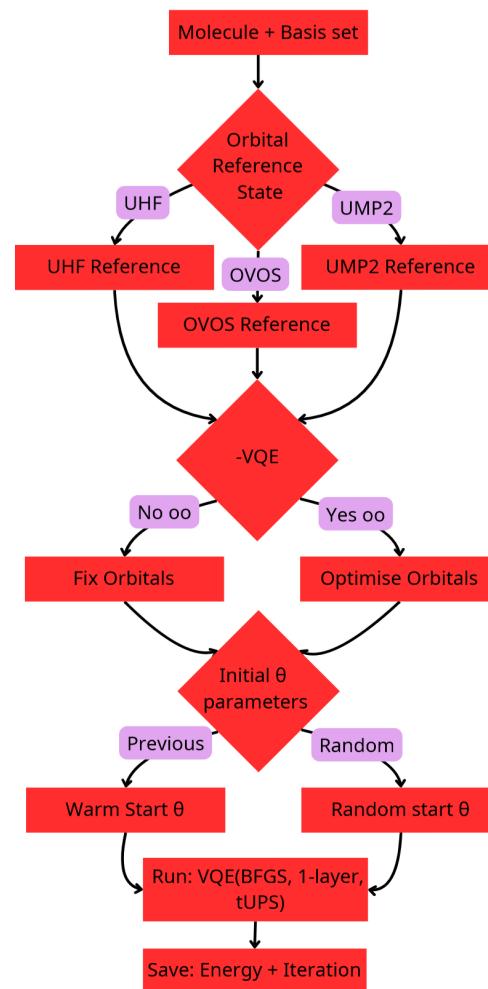
Key Takeaway: The first 50-68% of orbitals capture the bulk of the correlation.
All converged orbitals are unrestricted.

VQE

- **Goal:** Assess how OVOS orbitals improve VQE performance.



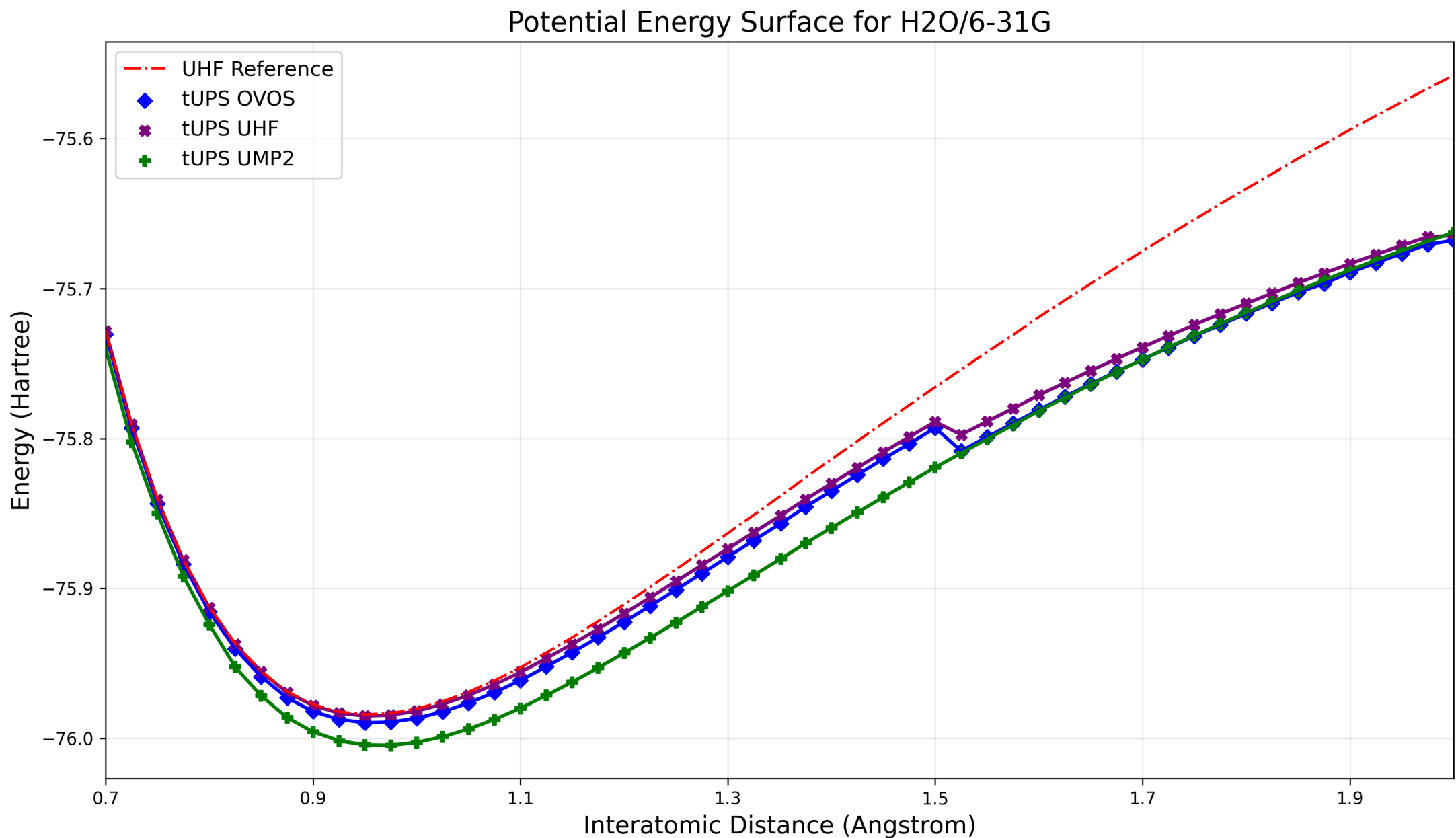
VQE workflow

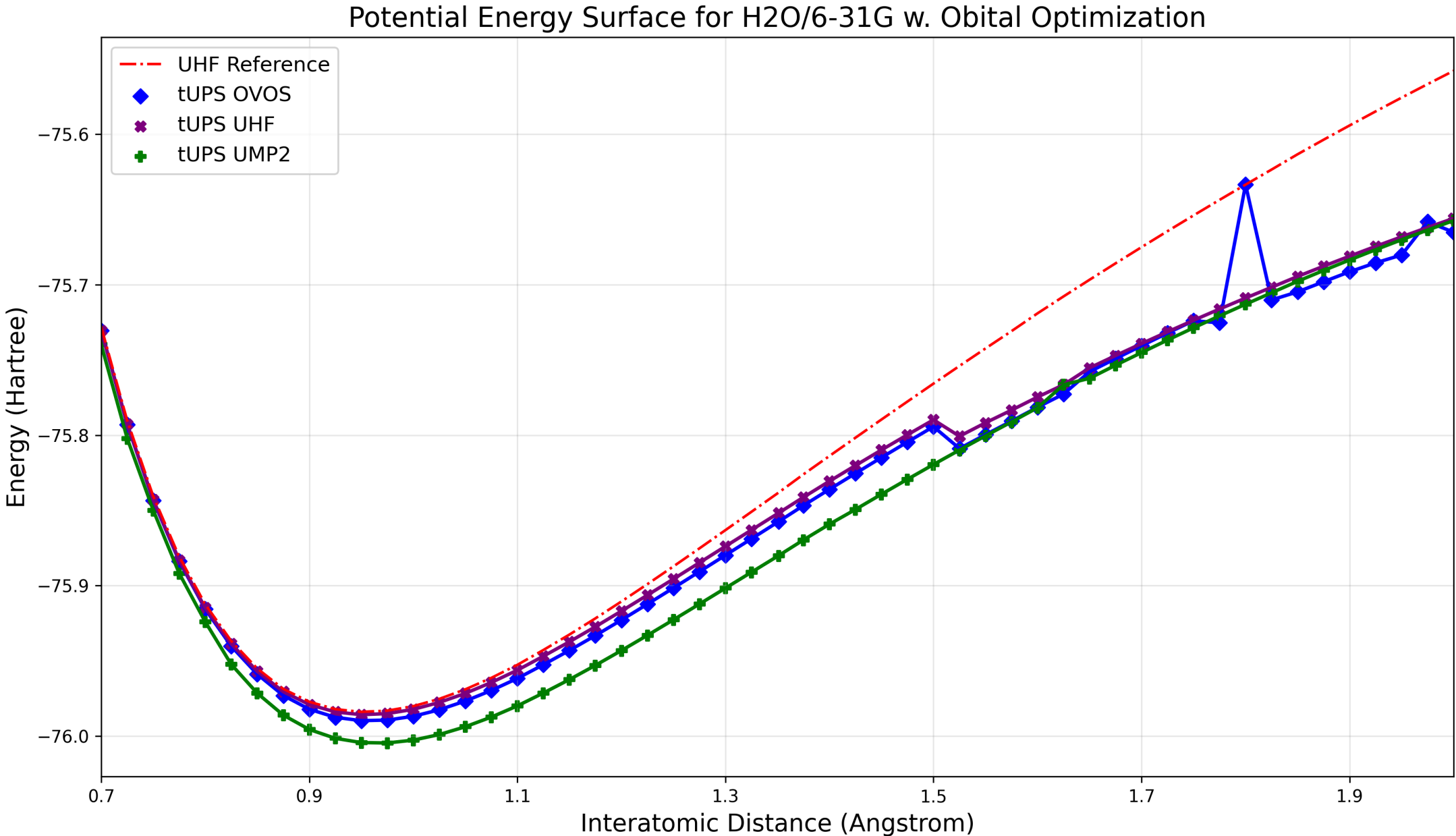


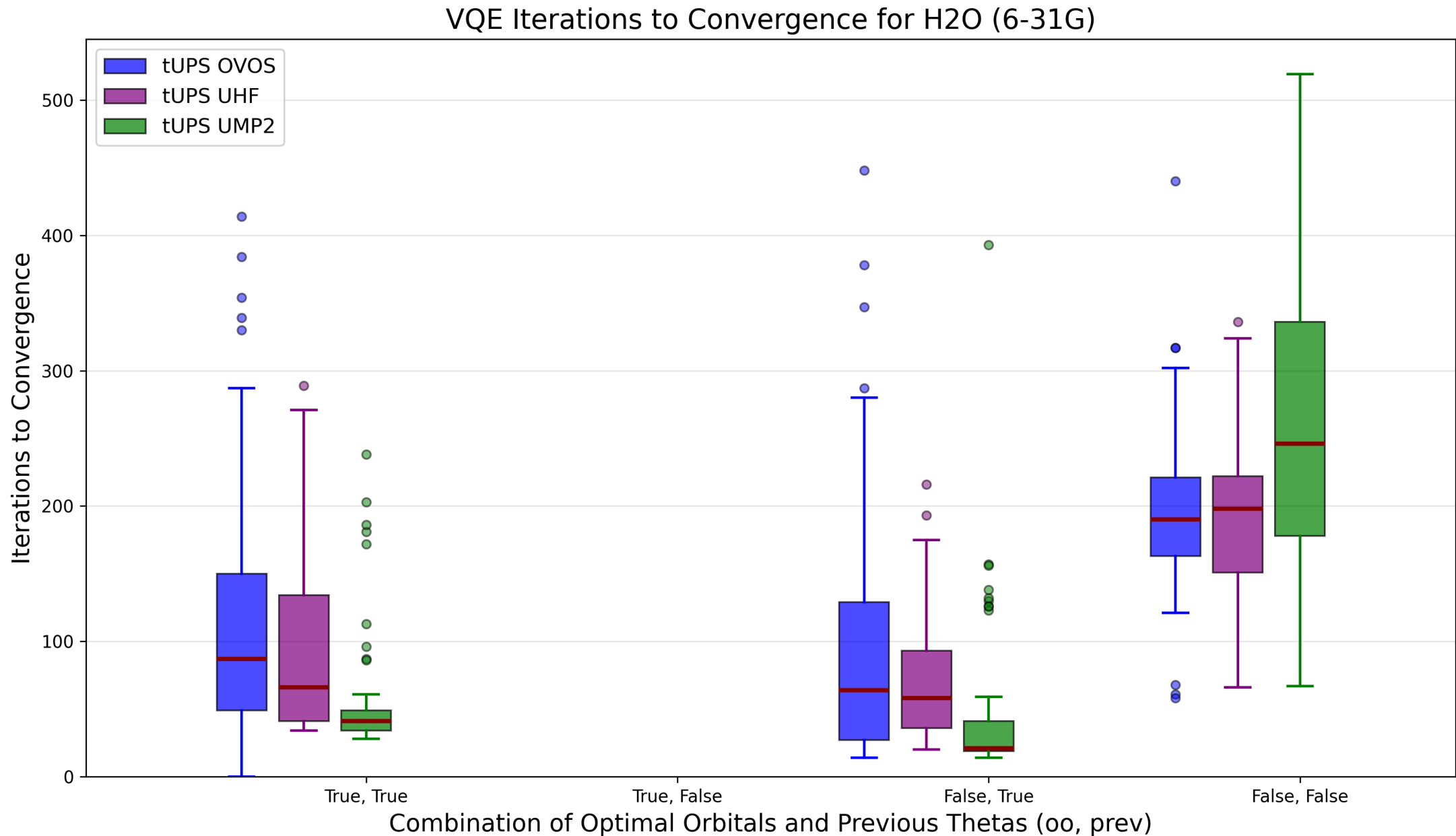
Results: VQE/ooVQE

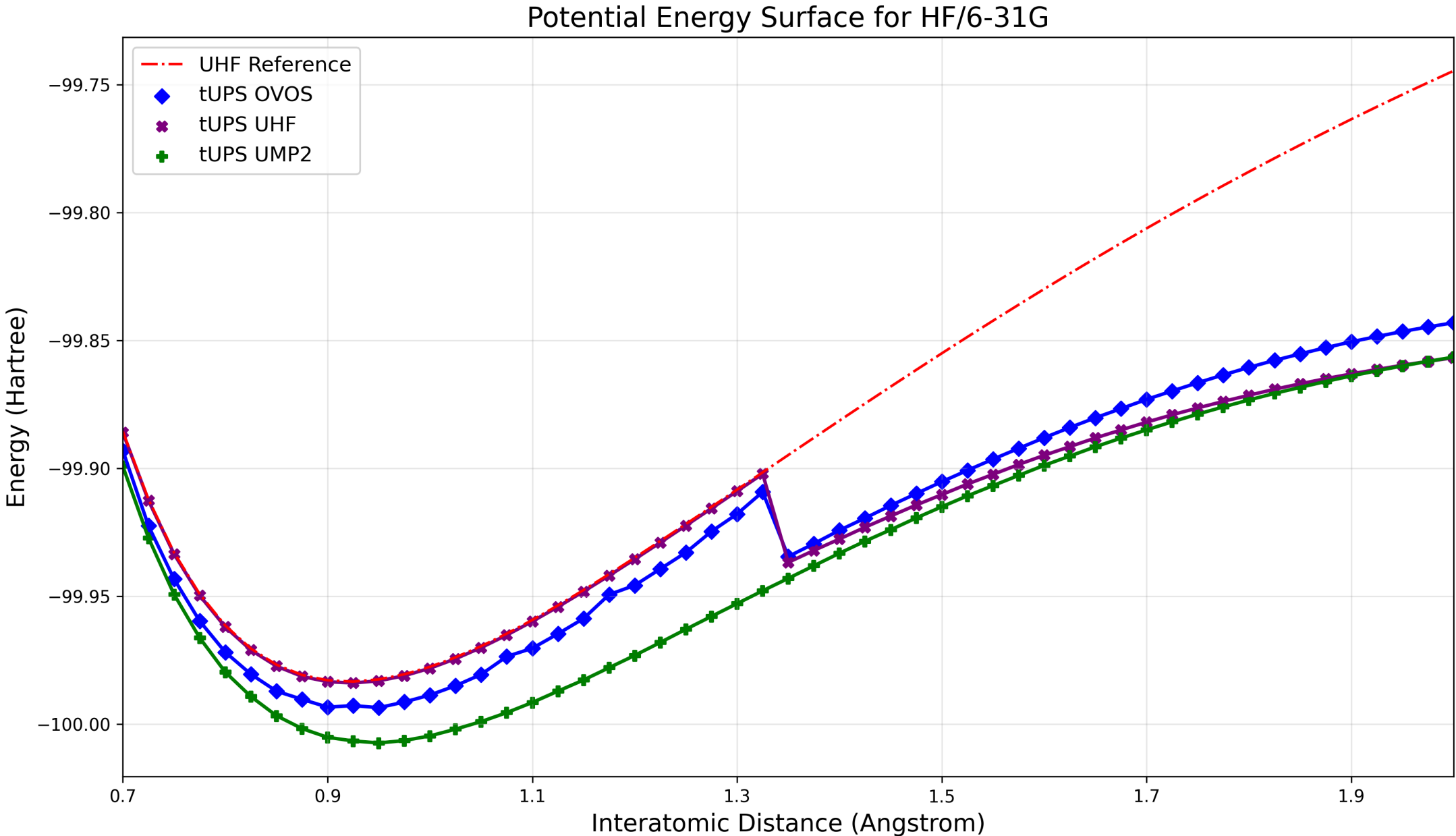
VQE plots

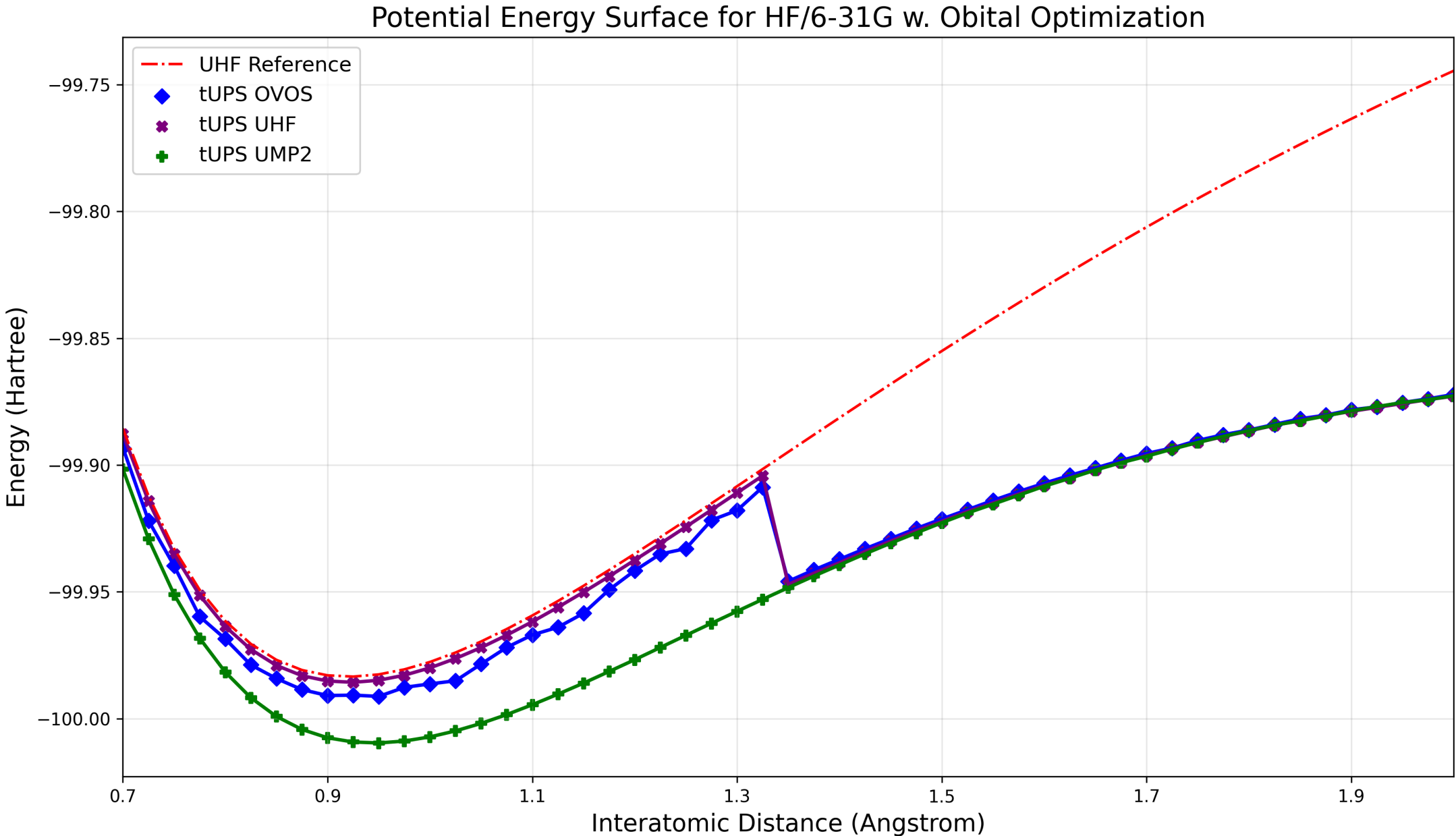
- *Points*: VQE energy at each iteration (Previous and Random).
- *Lines*: VQE energy convergence from initial point to final OVOS solution (UHF OVOS, UHF, and UMP2).
- *Dashed Line*: Reference UHF VQE.
- *ooVQE*: VQE with orbital optimisation of the virtual space.

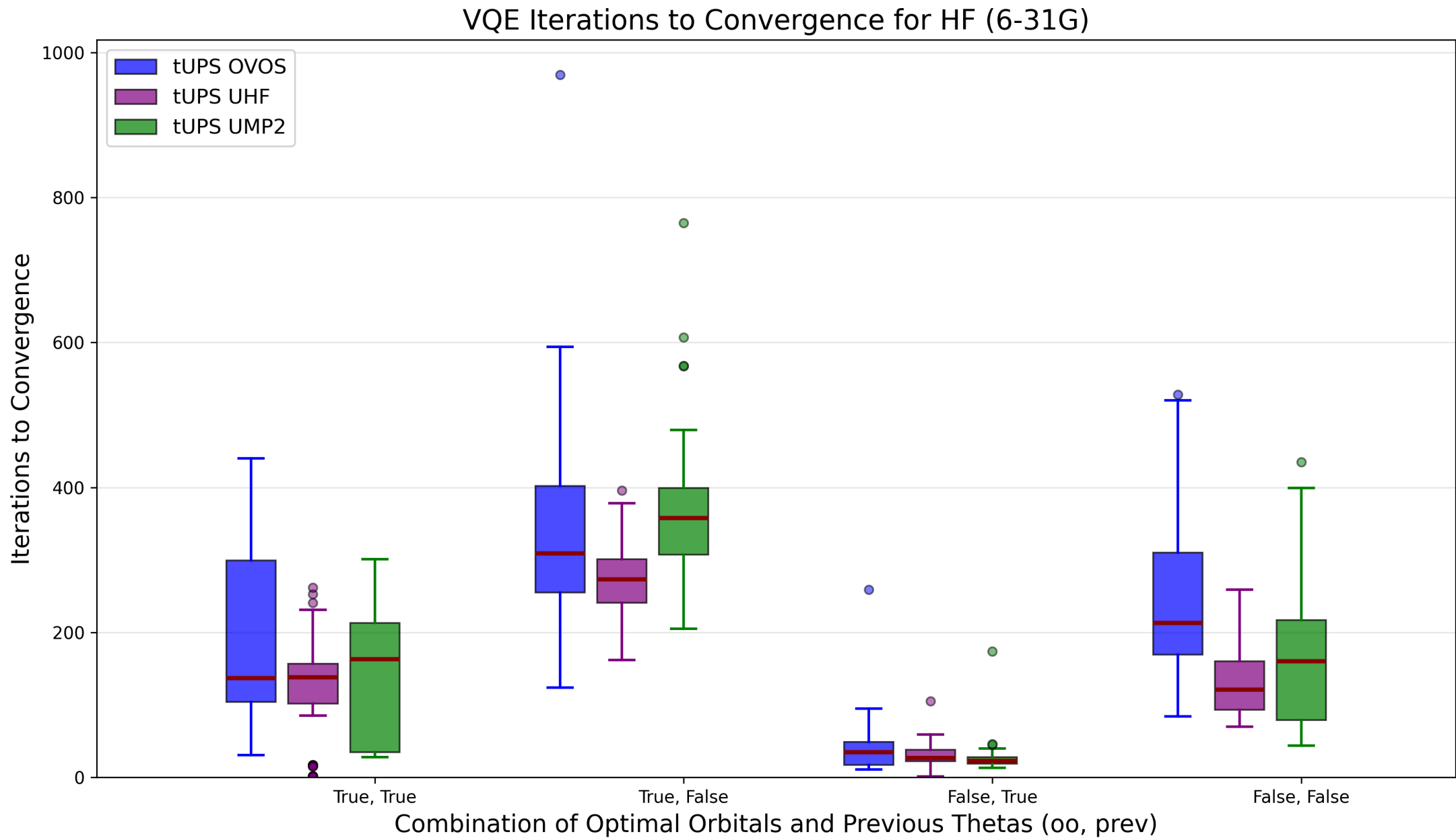


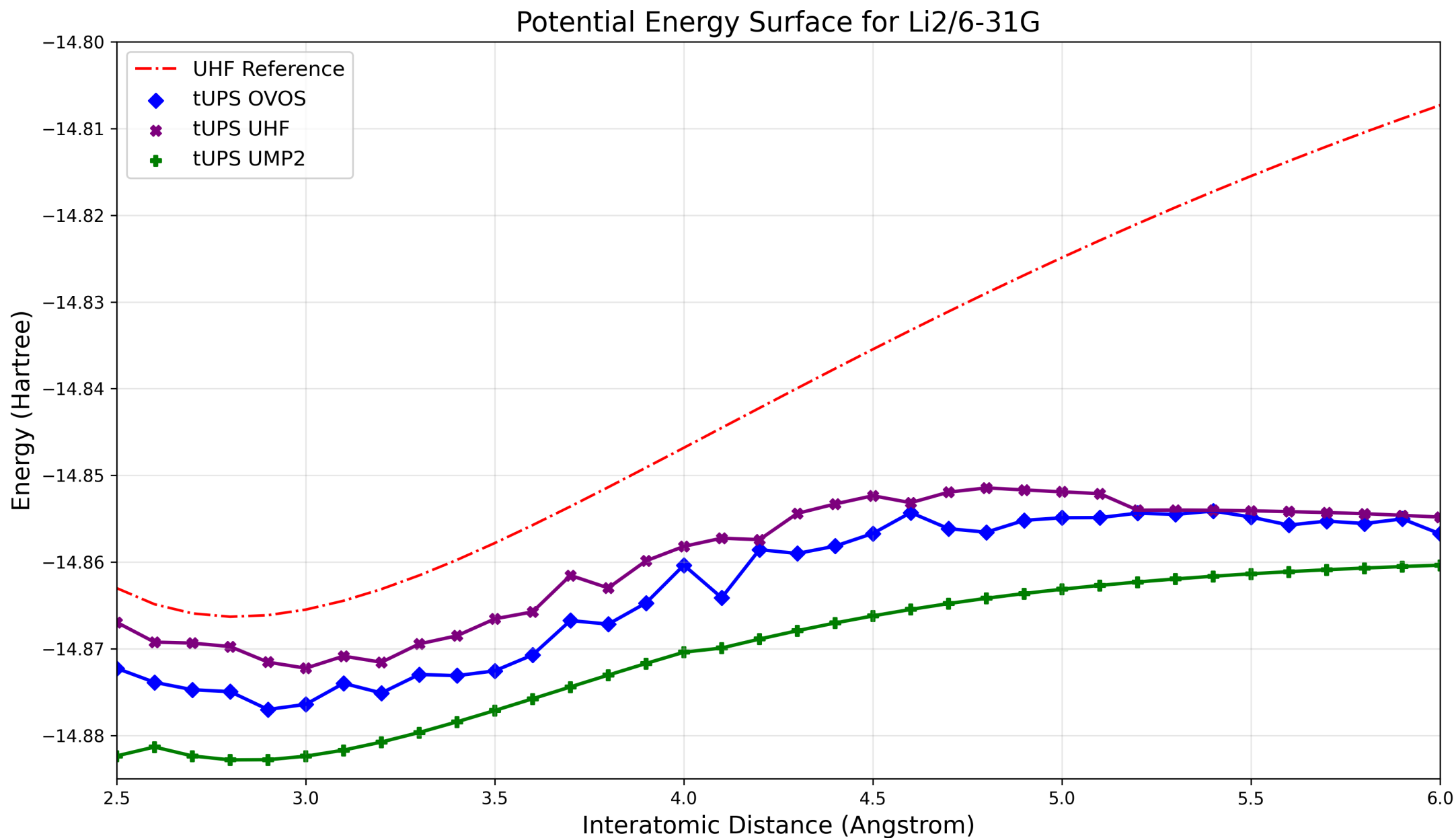


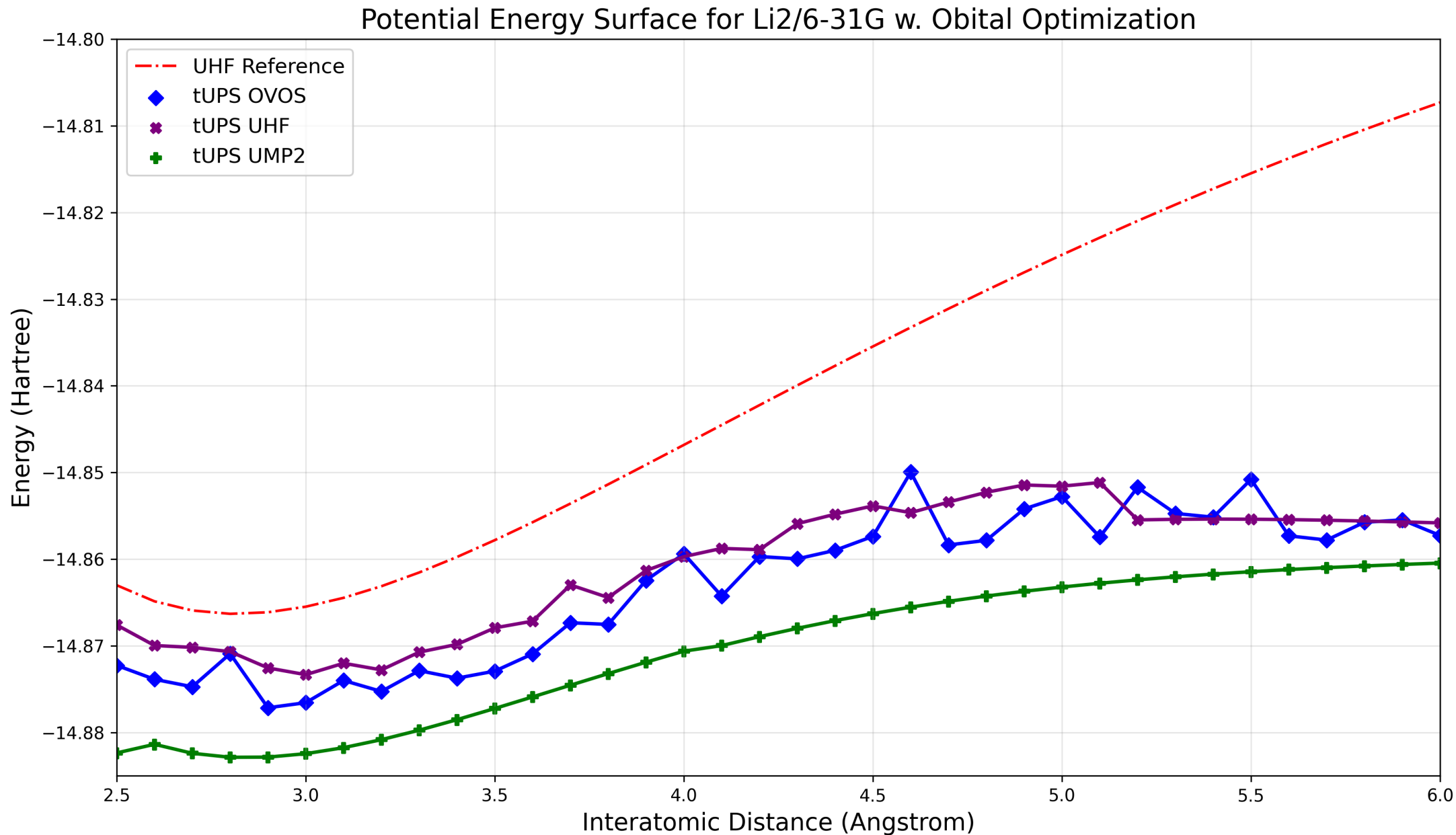


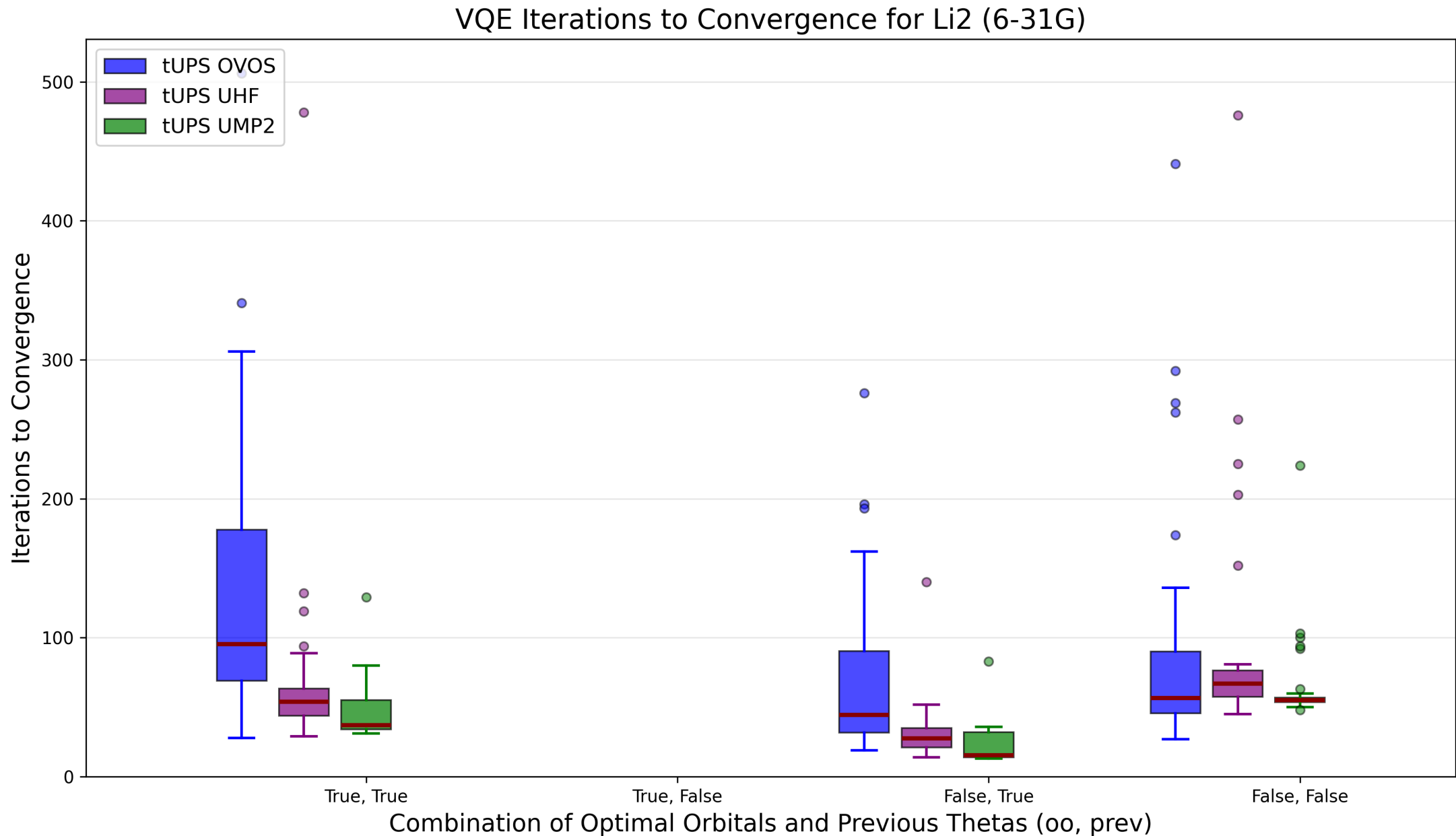












Results: VQE Summarized

- **VQE Bottleneck:** Canonical HF is a poor trial state.
- **OVOS Advantage:** Provides a superior trial state → better groundstate, lower energy convergence.
- **OVOS Disadvantage:** Provides an inferior trial state → worse groundstate, higher energy convergence.
- **Metric:** Reduced mean iterations for ooVQE.
- **Conclusion:** Classical optimisation of virtual orbitals directly reduces quantum circuit depth.



Conclusion

- ✓ **Successful implementation** of OVOS.
- ✓ **~90% MP2 correlation** recovered with **50–68% of virtual orbitals**.
- ✓ OVOS orbitals provide **superior VQE reference states** compared to UHF.
- ✓ Practical path to **reducing qubit count and circuit depth** for NISQ chemistry.



Outlook

Algorithmic improvements:

- Initial choice of virtual orbitals
- Trust-region / line-search for Newton-Raphson
- GPU acceleration, level-shifting for indefinite Hessian

Method extensions:

- OVOS for CCSD(T) and RHF reference
- Combine with natural orbital truncation

Quantum computing:

- Run VQE on quantum hardware using OVOS orbitals
- Explore reduced T1 $\rightarrow$ UCCD ansatz, shorter circuits

Larger systems:

- Transition metals
- Periodic systems
- Basis-sets



Thank You!

Questions?

